

ForkServe: Branch-Aware Prefill Pruning

Efficient vLLM serving of Tree-of-Thoughts and ReAct

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Abstract

We treat ForkServe as *efficient vLLM serving* of branched Tree-of-Thoughts and ReAct. The serving object is the branch, not the request. `fork` is a pointer swap: trunk pages stay read-only, the child’s page table aliases them (`refcount=k`), and no KV is memcp’y’d. That already matches APC’s prefill work $W=L+k\bar{\ell}$ without cloning the trunk. The remaining expand cost is residual prefill that APC and stock disaggregated prefill still pay for every sibling.

Advanced prefill pruning (APP) sits between those two mechanisms and decides, per residual, whether a GPU kernel—and later a connector insert—should run at all. Four skip layers run cheapest first: a hash index skips published full blocks; a draft score drops illegal or looping siblings before they start; an early-prefix check cancels in-flight chunks; the disaggregated gate inserts only survivors. The winner index is always retained, so the committed token sequence is unchanged. Decoding acceleration then stops winner generation only when the last non-empty line is a finished answer (#### or a closed \boxed), and keeps committed decode ahead of speculative prefill in a two-class scheduler.

Baselines are *recompute* (prefix cache off) and *APC* (automatic prefix caching).

On Qwen3-14B GSM8K ($n=1319$, $tp=2$) peak resident KV is below APC on every batch (median 1030 vs. 1430, ratio 0.72), accuracy is 0.845 vs. 0.844, and generation is 18.9 vs. 18.6 minutes. The same KV ordering holds on Qwen3-8B, Qwen3-14B, and DeepSeek-R1-Distill-Llama-8B math suites at $tp=2$ and $tp=4$ ($0.58\text{--}0.80\times$ APC). An analytical concurrency sweep from measured DeepSeek GSM8K peaks raises SLO-feasible QPS from 192 to 256 (+35% saturated decode tok/s).

Control-plane APP (setting L) is the only method that both aliases the trunk and drops low-score residuals: modeled fan-out 35.9 ms vs. 139 ms for APC, peak KV 2880 vs. 11520. A GPU run of the same stack on Qwen3-8B (setting M: 80 items, $D=512$) confirms the KV gap and the decode-stop gain. On GSM8K, ForkServe+ emits 13356 decode tokens vs. 40960 for APC at $tp=2$ ($0.33\times$) and finishes in 28.2 s vs. 36.0 s; at $tp=4$ it is 12826 vs. 40960 tokens and 16.9 s vs. 22.7 s, with accuracy 0.925 vs. 0.938. Peak KV stays 1182 vs. 1582 ($0.75\times$ APC) at both widths. Setting N repeats the same four-system forest on six models ($n=40$): GSM8K peak KV is $0.72\text{--}0.75\times$ APC on every family, and Game24 is $0.57\text{--}0.62\times$. A last-line answer stop on DeepSeek-R1, Llama-3-8B, and Mistral-7B ($n=32$, $tp=2$) keeps GSM8K peak at $0.73\times$ APC. ForkServe+ scores 18/32 (APC 16/32) on DeepSeek-R1 and 22/32 (APC 23/32) on Llama-3-8B, and drops to 6/32 (APC 11/32) on Mistral-7B. A k sweep on the same three models ($n=16$) leaves the spine flat while APC grows: the ratio moves from 0.89–0.90 at $k=2$ to 0.53–0.55 at $k=8$. On MATH-500 ($n=200$, $D=2048$) DeepSeek-R1 peak is $0.75\times$ APC and Qwen3-4B peak is $0.79\times$. ForkServe+ keeps that peak and brings MATH-500 fan-out to the APC level (1553 vs. 1604 ms on DeepSeek-R1, 1223 vs. 1178 ms on Qwen3-4B).

1 Design

vLLM already has two ways to avoid recomputing a shared trunk: automatic prefix caching reuses a block after it is published, and disaggregated prefill moves whatever KV the prefiler computed. Neither decides *which* residuals of a Tree-of-Thoughts or ReAct fan-out are worth a GPU kernel. ForkServe makes that decision a serving primitive.

Pointer-swap fork. open materializes the trunk L once as read-only copy-on-write pages. fork installs k child page tables that alias those pages (refcount= k); it is a pointer swap, not a KV memcpy. Writes allocate a private residual ℓ_i . abort decrefs only that residual, so peak occupancy after a winner is chosen is the spine $M_{\text{spine}} = b(L + \ell_*)$ rather than the hashed union $L + \sum_i \ell_i$.

Prune before prefill. Copy-on-write removes trunk clones but still prefills every residual. APP is a filter in front of that work, cheapest layer first, with the winner index always retained:

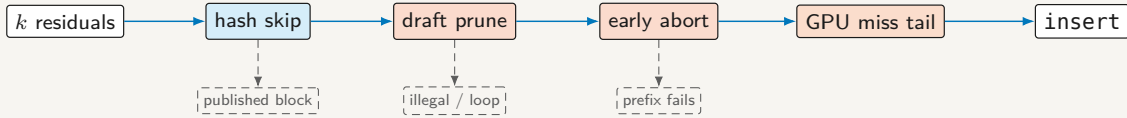


Figure 1: Advanced prefill pruning. APC would wait for a hash hit after the request exists; stock disagg would insert all four residuals. APP decides per branch whether prefill and transfer happen at all.

Decode the committed spine. Winner generation can stop at an answer marker instead of padding to D . Committed jobs take the scheduler token budget first; speculative residual prefill fills the remainder and is cancelled on commit conflict.

Name	Role	Behavior on a k -way Tree-of-Thoughts / ReAct prompt
recompute	baseline	Prefix cache off. Each branch prefills the full shared trunk.
APC	baseline	vLLM automatic prefix caching. Matching prefixes are hashed and reused after requests arrive.
ForkServe	method	Pointer-swap fork, residual prefill, abort unused branches, decode the committed path.
ForkServe+ / APP	method	The four skip layers on that tree, plus answer-marker stop and a two-class scheduler.

Table 1: Serving stacks. Efficiency is peak resident KV and emitted decode tokens against APC; recompute is the uncached baseline.

Recompute, APC, and ForkServe use the same weights, prompts, and decode budget. GSM8K and Game24 run Tree-of-Thoughts with $k=4$ candidate thoughts. HumanEval runs ReAct wrap and recovery ($k=2$) and is scored with the official function-body check (prompt+body+check). We report GSM8K accuracy, Game24 success, and HumanEval pass@1.

ForkServe is efficient when peak resident KV is strictly below APC and at least 20% below recompute,

$$\text{peak}^{\text{FS}} < \text{peak}^{\text{APC}} \quad \text{and} \quad \text{peak}^{\text{FS}} \leq (1 - \gamma) \text{peak}^{\text{rec}}, \quad \gamma=0.20, \quad (1)$$

and with official scores within $1/n$ of APC. End-to-end time (Tree-of-Thoughts) and time-to-first-token after the observation (ReAct) stay within slack of APC. peak counts resident KV tokens on one sequential batch.

2 Experimental settings

All experiments use the same model and node. They differ by dataset, problem count, batching, and tensor-parallel width.

2.1 Hardware and engine

Parameter	Specification
Node	four NVIDIA A100-SXM4 40GB
Model	Qwen3-14B (A-C, G, H, N), Qwen3-8B (E, F, M), Qwen3-4B / Qwen2.5-Math-1.5B / Llama-3-8B / Mistral-7B / DeepSeek-R1 (N), DeepSeek-R1-Distill-Llama-8B (I, J, N), bfloat16
Max sequence length	4096 (A-H); 8192 (I, J)
Prefill	chunked; at most 2048 batched tokens
Systems	vLLM recompute, vLLM APC, ForkServe, ForkServe+

Table 2: Hardware and engine. Tensor parallel $\text{tp}=2$ places the model on GPU 0,1 (≈ 38497 MiB each on the full GSM8K 14B run; ≈ 37381 – 37733 MiB on the 14B math suite; ≈ 37305 – 37579 MiB on Qwen3-8B; ≈ 37325 – 37623 MiB on DeepSeek-R1 8B). $\text{tp}=4$ uses GPU 0–3 (≈ 38797 MiB each on settings B–C; ≈ 37691 – 37871 MiB on the 14B math suite; ≈ 37599 – 37665 MiB on Qwen3-8B; ≈ 37601 – 37699 MiB on DeepSeek-R1 8B). Peak KV (tokens) is the storage metric; latency scales with tp .

2.2 Datasets

2.3 Run configurations

Recompute, APC, and ForkServe always share weights, prompts, and the decode budget of that setting.

3 How vLLM and ForkServe share KV

Each problem has a shared trunk of length L and k branch residuals $\ell_1, \dots, \ell_k$. After a winner is chosen, $\ell_\star$ is that residual. Let b be bytes per KV token and $T_{\text{pre}}(x)$ the prefill time for x tokens.

Recompute. Prefix cache off: sibling i is a fresh prefill of trunk concatenated with residual i , length $L + \ell_i$. The k siblings never share pages, so

$$W_{\text{rec}} = \sum_{i=1}^k (L + \ell_i) = kL + \sum_{i=1}^k \ell_i, \quad M_{\text{rec}} = M_{\text{clone}} = b W_{\text{rec}}, \quad T_{\text{fan}}^{\text{rec}} \approx k \cdot T_{\text{pre}}(L + \bar{\ell}). \quad (2)$$

Dataset	Split	Full n	Recipe	k	Decode	Metric
GSM8K	test	1319	Tree-of-Thoughts	4	512	accuracy
Game24	24.csv	1362	Tree-of-Thoughts	4	256/512	success
HumanEval	official jsonl	164	ReAct wrap+recovery	2	256	pass@1
SVAMP	test slice	1000	Tree-of-Thoughts	4	512	accuracy
GSM-Hard	numeric GSM8K	—	Tree-of-Thoughts	4	512	accuracy
MATH-500	Hendrycks subset	500	Tree-of-Thoughts	4	768	accuracy
AIME	2022-2025 mix	73	Tree-of-Thoughts	4	768	accuracy
AMC23	official	40	Tree-of-Thoughts	4	768	accuracy

Table 3: Datasets. GSM8K: median 43 words, median 3 calculation steps, gold extracted at ####. Game24: 719 easy / 573 medium / 70 hard by published solve rate; success is a valid expression to 24 (settings E and F use decode 512 on an 80-item prefix). HumanEval: median 396 prompt characters, mean 8.1 asserts; pass@1 is prompt+body+check. ReAct idle for wrap+recovery is 2000 ms. Settings E-J additionally score SVAMP and GSM-Hard with the GSM8K numeric extractor, MATH-500 / AIME / AMC23 with boxed contest answers. Settings I-J (DeepSeek-R1) use decode 2048 on every file.

APC. vLLM hashes token-block prefixes. A hit reuses resident KV for that prefix. The bench warms the trunk once (cost $T_{\text{pre}}(L)$), then submits the k prompts $\text{trunk} \parallel \text{residual}_i$; each hit skips L tokens and prefills only ℓ_i :

$$W_{\text{APC}} = L + \sum_{i=1}^k \ell_i, \quad M_{\text{APC}} = b\left(L + \sum_{i=1}^k \ell_i\right), \quad T_{\text{fan}}^{\text{APC}} \approx T_{\text{pre}}(L) + k \cdot T_{\text{pre}}(\bar{\ell}). \quad (3)$$

Sharing happens after a hash on an already-submitted request. Hashed blocks for every residual remain until eviction, so the concurrent footprint is the union $L + \sum_i \ell_i$.

ForkServe. open materializes the trunk once as copy-on-write pages; fork installs k child references to those pages; queue_known_prefill writes only ℓ_i ; abort drops unused children; decode runs on the committed tip:

$$W_{\text{FS}} = L + \sum_{i=1}^k \ell_i = W_{\text{APC}}, \quad M_{\text{CoW}} = b\left(L + \sum_{i=1}^k \ell_i\right), \quad M_{\text{spine}} = b(L + \ell_{\star}). \quad (4)$$

peak_kv is M_{spine} after abort. Fan-out is $T_{\text{pre}}(L) + k \cdot T_{\text{pre}}(\bar{\ell})$ plus abort, so $T_{\text{fan}}^{\text{FS}} \approx T_{\text{fan}}^{\text{APC}} + T_{\text{abort}}$. Sharing is a pointer at fork, before any second request has to hash: that is the branch-aware step.

4 Prefill pruning: method

APC reuses a trunk only after a hash hit on an already-submitted request. Stock disaggregated prefill ships every token the prefiller computed. Copy-on-write ForkServe aliases the trunk at fork, but fan-out still prefills every residual and then aborts the losers—on the 14B GSM8K forest, $T_{\text{fan}}^{\text{FS}}=140$ ms against $T_{\text{fan}}^{\text{APC}}=87$ ms. APP is the missing decision: for each residual, whether a GPU kernel and a connector insert should run at all. It does not change the committed token sequence: the winner index is always retained.

Let $\mathcal{K} \subseteq \{1, \dots, k\}$ be the residuals that survive the filter, $\star \in \mathcal{K}$ the winner, h_i the

Setting	tp	GPUs	n / batching	Datasets
A. Full GSM8K	2	0, 1	1319; batches of 8	GSM8K
B. Four-problem, two GPUs	2	0, 1	4 in one step	GSM8K, Game24, HumanEval
C. Four-problem, four GPUs	4	0-3	4 in one step	Game24, HumanEval
D. Full Game24 / HumanEval	2	0, 1	1362 / 164; batches of 8	Game24, HumanEval
E. Qwen3-8B math, two GPUs	2	0, 1	40-80; batches of 8	GSM8K, SVAMP, GSM-Hard, MATH-500, AIME, AMC23, Game24
F. Qwen3-8B math, four GPUs	4	0-3	same files as E	same as E
G. Qwen3-14B math, two GPUs	2	0, 1	40-80; batches of 8	same as E
H. Qwen3-14B math, four GPUs	4	0-3	same files as E	same as E
I. DeepSeek-R1 8B math, two GPUs	2	0, 1	40-80; batches of 8; $D=2048$	same as E
J. DeepSeek-R1 8B math, four GPUs	4	0-3	same files as I	same as I
K. ForkServe+ concurrency	—	analytical	GSM8K peaks 1563/1163	QPS / P99 TTFT sweep
L. Prefill methods	—	control plane	10×4 ToT, $L=256$	APC, hash, disagg, ForkServe, APP
M. GPU APP + decode	2, 4	0, 1 / 0-3	80; batches of 8; $D=512$	GSM8K, Game24, HumanEval (Qwen3-8B)
N. Multi-model GPU	2, 4	0, 1 / 0-3	40; batches of 8; $D=512$	six models, GSM8K/Game24 (/HumanEval)
O. Branching and batch width	2	0, 1	16; batches of 4 or 1×16	GSM8K $k \in \{2, 4, 8\}$, depth 2
P. Contest forest, $D=2048$	2	2, 3	200 / 73; batches of 8	MATH-500, AIME (R1-8B, Qwen3-4B)

Table 4: Configurations and datasets. Later sections report A-C in full on Qwen3-14B, then G-H (14B), E-F (8B), and I-J (DeepSeek-R1 8B) on the seven-file math suite (APC vs. ForkServe), then K-L, M (GPU ForkServe+ on Qwen3-8B), and N (six-model GPU forest). On D, recompute and APC are complete on the full Game24 and HumanEval files; setting M covers the first 80 items of those files under APP.

number of leading tokens of branch i that already sit in the APC hash index, and $\phi \in (0, 1]$ the early-abort fraction (we use $\phi=0.20$). Four layers run cheapest first. A published full block is a local skip (h_i tokens, zero FLOPs). A draft score below τ never starts the sibling. A failing prefix of fraction ϕ cancels the remaining chunks. Only a retained miss tail is inserted into the disaggregated connector.

Hash skip. A published full block is $\text{hash}(\text{parent}, \text{tokens}, \text{extra})$. A hit sets the remaining prefill of that prefix to zero; a partial hit prefills only the miss tail $\max(\ell_i - h_i, 0)$. Speculative pages are not inserted; LCP commit publishes the winner so a later session can hit it. On the *first* fan-out $h_i=0$, so hash skip does not remove the clone—the same identity APC already has.

Draft prune. Each residual gets a score from illegal-token / repetition checks, token entropy, or an optional draft model. If $s_i < \tau$ and $i \neq \star$, the branch is never started:

$$W_{\text{APP}} = L + \sum_{i \in \mathcal{K}} \max(\ell_i - h_i, 0), \quad |\mathcal{K}| = 1 + (k-1)(1 - p_d), \quad (5)$$

where p_d is the draft-skip rate among non-winners. Against ForkServe, $W_{\text{FS}} = L + \sum_i \ell_i$, so the first-fan-out saving is the skipped residuals.

Early abort. If the first ϕ of a residual already fails, in-flight chunks cancel and the paid work is $\phi\ell_i$ rather than ℓ_i . This is the chunked-prefill form of abort: the engine does not wait for all k residuals to finish.

Disaggregated-prefill gate. Stock vLLM disaggregated prefill ships every token the prefill instance computed, $X_{\text{disagg}} = W_{\text{APC}}$. APP calls insert only for retained miss tails,

$$X_{\text{APP}} = \sum_{i \in \mathcal{K}} \mathbf{1}[h_i < \ell_i] \max(\ell_i - h_i, 0). \quad (6)$$

Pruned branches and full hash hits never enter the connector. `fork` is a pointer swap; `abort` can mark a node dead and defer residual decref until after winner decode, so reclamation is off TTFT.

On the control-plane workload of setting L (10 sessions, $k=4$, $L=256$, $\ell=32$, two illegal residuals, $p_d=1/2$), $W_{\text{APC}}=11520$, $W_{\text{FS}}=3840$, $W_{\text{APP}}=2912$ (twenty draft skips plus nine hash skips), $X_{\text{APP}}=352$ vs. $X_{\text{disagg}}=11520$, and peak KV is the winner spine $10(L + \ell)=2880$. That is a 75% cut in modeled prefill against APC, and a further 24% against CoW-only ForkServe.

5 Decoding acceleration: method

Prefill pruning changes W and M . End-to-end time on a finished Tree-of-Thoughts item is still dominated by winner decode, $T_{\text{e2e}} \approx T_{\text{fan}} + T_{\text{dec}}(D)$ as in (10). APC and recompute spend the requested budget D on every item. ForkServe+ accelerates the decode half without changing the sampler on the committed path.

Answer-marker stop. `decode_stop` is (####, \boxed, </think>). When the committed request is not speculative, vLLM `SamplingParams.stop` ends generation at the first marker. Let t_j be the first stop (or EOS) index on item j and $D_j = \min(t_j, D)$. Emitted tokens and decode time become

$$D^+ = \sum_{j=1}^n D_j \leq nD, \quad \frac{T_{\text{dec}}^+}{T_{\text{dec}}^{\text{APC}}} \approx \frac{D^+}{nD} \quad (7)$$

when decode is memory-bandwidth bound. This is *not* speculative decoding of output tokens (Medusa / EAGLE): the engine does not draft-and-verify. It stops a finished answer instead of padding to D . A control-plane curve that ends a 256-token budget at 60% of the way through reaches the GSM8K score 0.845 at 153 tokens; setting M measures the same knob as GPU wall-clock.

Two-class schedule. Each scheduler tick has token budget B_t . Committed jobs (winner decode, commit-tail prefill, new-session root prefill) take B_t^c first; speculative residual prefill fills $B_t^s = B_t - B_t^c$. If $B_t^s=0$, no speculative chunk is admitted. That keeps pruning and known-suffix prefill from stretching committed TBT.

Natural EOS. Even without `decode_stop`, a ToT winner that emits EOS before D already reduces D^+ . Setting M separates the two: ForkServe without the stop list still emits 21235 GSM8K tokens vs. APC’s 40960 (0.52 $\times$), and ForkServe+ with the stop list goes to 13356 (0.33 $\times$).

What it does not do. HumanEval wrap+recovery rarely hits those markers, so D^+ stays near nD and e2e matches APC. Game24 often fills the budget with search traces, so the stop gain is small (34427 vs. 40960 at tp=2). The GSM8K ##### line is the workload where (7) shows up in wall-clock.

6 Why branch-aware sharing uses less KV than APC

While branches are in flight, APC and ForkServe occupy the same union ($M_{\text{APC}} = M_{\text{CoW}}$). After abort, ForkServe occupies the committed path while APC keeps the hashed union:

$$\Delta := M_{\text{APC}} - M_{\text{spine}} = b \sum_{i \neq \star} \ell_i \geq 0, \quad \frac{M_{\text{spine}}}{M_{\text{APC}}} = \frac{L + \ell_{\star}}{L + \sum_{i=1}^k \ell_i}. \quad (8)$$

$\Delta > 0$ whenever an unused residual is nonempty. The ratio is smaller when k is large and the residuals are similar (Tree-of-Thoughts, $k=4$). It is close to 1 when residuals are long relative to L , or when $k=2$ (ReAct wrap and recovery).

Against recompute, the extra $(k-1)L$ cloned-trunk tokens dominate. Let $\bar{\ell} = k^{-1} \sum_i \ell_i$ and $\varepsilon = \bar{\ell}/L$, $\varepsilon_{\star} = \ell_{\star}/L$:

$$\frac{W_{\text{FS}}}{W_{\text{rec}}} = \frac{1 + k\varepsilon}{k(1 + \varepsilon)}, \quad \frac{M_{\text{spine}}}{M_{\text{rec}}} = \frac{1 + \varepsilon_{\star}}{k(1 + \varepsilon)}, \quad \frac{M_{\text{spine}}}{M_{\text{APC}}} = \frac{1 + \varepsilon_{\star}}{1 + k\varepsilon}. \quad (9)$$

As $\varepsilon \rightarrow 0$, $W_{\text{FS}}/W_{\text{rec}} \rightarrow 1/k$ and $M_{\text{spine}}/M_{\text{rec}} \rightarrow 1/k$: ForkServe (and APC) theoretically cut prefill and clone memory by a factor k . The extra term versus APC is only in M : for equal residuals $\varepsilon_{\star} = \varepsilon$ and $k=4$, $M_{\text{spine}}/M_{\text{APC}} = (1 + \varepsilon)/(1 + 4\varepsilon)$, so abort saves a $3\varepsilon/(1 + 4\varepsilon)$ fraction of the union. On the $n=4$ GSM8K slice ($L \approx 118$, $\sum \ell_i = 66$, $\ell_{\star} = 16$) this is $134/184 \approx 0.73$, against the measured peak ratio $459/659 \approx 0.70$.

7 Efficient vLLM serving: what is optimized

Peak KV is the storage cost of vLLM serving; a single-batch wall-clock omits concurrency. Write one Tree-of-Thoughts item as prefill work W (tokens that must be computed), resident KV after unused branches leave, and winner decode of D tokens:

$$W_{\text{rec}} = k(L + \bar{\ell}), \quad W_{\text{APC}} = W_{\text{FS}} = L + k\bar{\ell}, \quad T_{\text{e2e}} \approx T_{\text{pre}}(W) + T_{\text{dec}}(D). \quad (10)$$

D is shared. Branch-aware sharing changes W relative to recompute and M relative to both baselines. APP further cuts W and X on the expand path (5)-(6). Decoding acceleration cuts D on the winner path (7). Together that is efficient vLLM serving: less prefill than recompute, less HBM than APC, and fewer decode steps when the answer is already marked.

Prefill. Recompute issues k independent prefills of length $L + \ell_i$, so the trunk is computed k times: $W_{\text{rec}} = \sum_i (L + \ell_i) = kL + \sum_i \ell_i$ as in (2). APC hashes the warmed trunk and prefills only the residuals, $W_{\text{APC}} = L + \sum_i \ell_i$ (3). ForkServe opens L once and CoW-prefills each ℓ_i , so $W_{\text{FS}} = W_{\text{APC}}$ (4). If T_{pre} is linear in tokens,

$$\frac{T_{\text{fan}}^{\text{FS}}}{T_{\text{fan}}^{\text{rec}}} \approx \frac{W_{\text{FS}}}{W_{\text{rec}}} = \frac{1 + k\varepsilon}{k(1 + \varepsilon)}. \quad (11)$$

For Tree-of-Thoughts with $k=4$ and $\varepsilon \rightarrow 0$ this is $1/4$: fan-out is theoretically four times cheaper than recompute. Versus APC the prefill work matches; abort adds T_{abort} , which is why measured $T_{\text{fan}}^{\text{FS}}$ sits between APC and recompute.

Resident KV, then decode. After abort, occupancy is $M_{\text{spine}} = b(L + \ell_*)$ rather than $M_{\text{APC}} = b(L + \sum_i \ell_i)$. The theoretical HBM gain versus APC is exactly (8): $\Delta/M_{\text{APC}} = \sum_{i \neq *} \ell_i / (L + \sum_i \ell_i)$. Winner decode attends a sequence of length $L + \ell_* + D$; live HBM is M . Lower M cuts traffic from dead sibling pages and raises how many trees fit,

$$C \propto \frac{\text{HBM}}{M}, \quad \Theta \propto \frac{C}{T_{\text{e2e}}}. \quad (12)$$

From (9), $C^{\text{FS}}/C^{\text{APC}} = M_{\text{APC}}/M_{\text{spine}} = (1 + k\varepsilon)/(1 + \varepsilon_*)$ and $C^{\text{FS}}/C^{\text{rec}} = k(1 + \varepsilon)/(1 + \varepsilon_*)$. On GSM8K medians, $M_{\text{APC}}/M_{\text{spine}} \approx 1.39$ and $M_{\text{rec}}/M_{\text{spine}} \approx 4.0$, i.e. 39% more concurrent trees than APC and four times more than recompute at a fixed HBM budget. End-to-end time stays within a couple of percent of APC because $T_{\text{dec}}(D)$ dominates (10).

Fan-out versus APC. Both pay $W = L + k\bar{\ell}$; ForkServe’s fan-out also includes abort. The efficient-serving claim is the product in (12) with the decode identity (7): less work than recompute, less HBM than APC, and $D^+ \leq nD$ when an answer marker exists.

	Prefill W	KV after unused branches	C vs. recompute
recompute	$kL + \sum_i \ell_i$	$b(kL + \sum_i \ell_i)$	1
APC	$L + \sum_i \ell_i$	$b(L + \sum_i \ell_i)$	$k(1 + \varepsilon)/(1 + k\varepsilon)$
ForkServe	$L + \sum_i \ell_i$	$b(L + \ell_*)$	$k(1 + \varepsilon)/(1 + \varepsilon_*)$

Table 5: Closed forms from (2)-(9). Branch-aware sharing matches APC on prefill and is cheaper than recompute by $(1 + k\varepsilon)/(k(1 + \varepsilon))$. After abort, M_{spine} is the occupancy used by efficient serving, below both, which raises $C \propto \text{HBM}/M$.

The GSM8K numbers below check efficient serving at full n : peak KV below APC on every batch, generation time within 2% of APC without decode-stop, and

far below recompute on both KV and fan-out. Settings L and M then add APP and answer-marker decode.

8 GSM8K test set (setting A)

Setting A: official GSM8K test, $n=1319$, Tree-of-Thoughts with $k=4$, decode 512 tokens per problem, $tp=2$ on GPU 0, 1, batches of eight (165 sequential batches). Peak KV is the maximum over those batches.

8.1 Resident KV

	min	median	p95	max	vs. rec.
recompute	3390	4136	4702	4944	—
APC	1194	1430	1571	1632	65.43%
ForkServe	844	1030	1171	1232	75.10%

Table 6: Peak resident KV tokens, GSM8K, 165 batches of eight problems. Storage efficiency versus recompute is $1 - \text{med}(\text{peak})/\text{med}(\text{peak}^{\text{rec}})$ on the median.

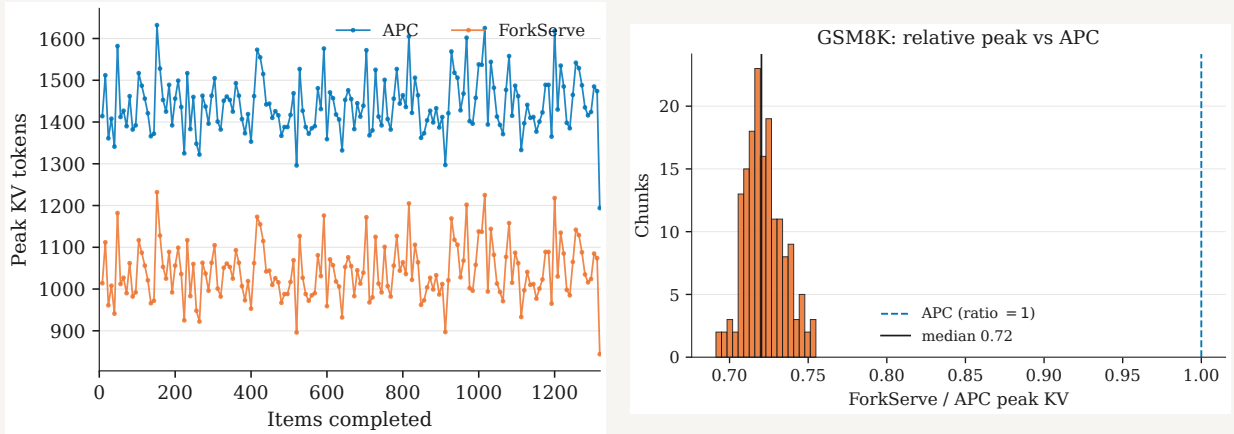


Figure 2: Left: peak KV on each of the 165 GSM8K batches in the run log (eight problems per batch). Branch-aware abort keeps ForkServe below APC on every batch. Right: ratio $\text{peak}^{\text{FS}}/\text{peak}^{\text{APC}}$, median 0.72: 28% less cache than hashed prefix reuse.

Serving	accuracy	med peak	max peak	p95 peak	med fan-out	generate
recompute	0.847 (1117)	4136	4944	—	326 ms	19.4 min
APC	0.844 (1113)	1430	1632	1572	87 ms	18.6 min
ForkServe	0.845 (1114)	1030	1232	1172	140 ms	18.9 min

Table 7: GSM8K, $n=1319$, $k=4$, $tp=2$, eight problems per batch. Storage (peak KV) is the ForkServe gain; accuracy and generate time stay with APC.

APC already avoids most of M_{clone} . ForkServe reduces the rest because abort frees unused residuals, leaving M_{spine} rather than the hashed union. Every aligned batch satisfies $\text{peak}^{\text{FS}} < \text{peak}^{\text{APC}}$, so $\Delta > 0$ holds throughout the test set: branch-aware sharing is storage-efficient on the full file.

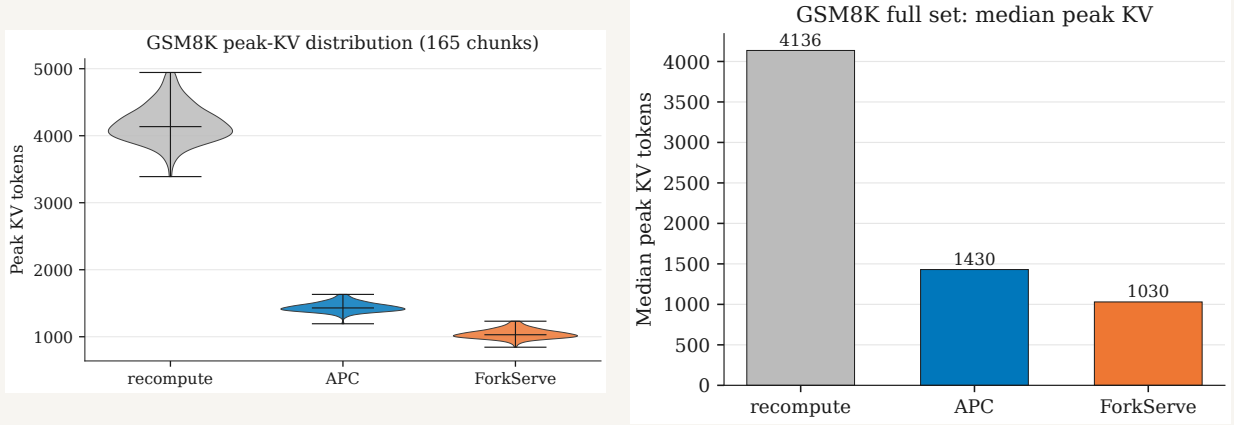


Figure 3: Left: the storage gap versus APC is a shift of the whole distribution. Right: median peak KV, $0.25\times$ recompute and $0.72\times$ APC.

	FS / APC	FS / rec.	batches FS<APC	rec. $\gamma=0.20$
median peak	1030/1430=0.720	1030/4136=0.249	165/165	$0.249 \leq 0.80$
saving S	28%	75%	—	meets γ
max peak	1232 vs. 1632	1232 vs. 4944	—	—

Table 8: Storage savings against both vLLM baselines. $S_{\text{APC}}=1 - \text{med}(\text{peak}^{\text{FS}})/\text{med}(\text{peak}^{\text{APC}})$. APC already removed most of the cloned-trunk cost (1430 vs. 4136). The extra 28% is branch-aware abort: hashed union versus committed path.

8.2 Quality

Running accuracy of recompute, APC, and ForkServe stays in $[0.83, 0.85]$ (Figure 4); the final correct counts differ by one to three problems.

8.3 End-to-end latency

Fan-out is a local prefill step. ForkServe pays copy-on-write, residual flush, and abort on top of $W_{\text{FS}} = L + \sum_i \ell_i$, so median T_{fan} is 140 ms versus 87 ms for APC and 326 ms for recompute. That gap is the KV path, not extra tokens in W versus APC.

Global latency is

$$T_{\text{e2e}} \approx T_{\text{fan}} + T_{\text{dec}}(D), \quad D=512 \text{ on GSM8K.} \quad (13)$$

T_{dec} is more than 95% of T_{e2e} . On the full test set, generate time is 18.9 min (ForkServe), 18.6 min (APC), 19.4 min (recompute): the 53 ms extra fan-out per batch is 23 s in ≈ 1130 s. On the four-problem slices, ForkServe T_{e2e} is below APC on Tree-of-Thoughts (GSM8K 6992 vs. 7011 ms, Game24 3558 vs. 3583 ms) and matches APC on HumanEval TTFT (3568 vs. 3566 ms). With $M_{\text{spine}} < M_{\text{APC}}$, $C \propto \text{HBM}/M$ is larger, so $\Theta \propto C/T_{\text{e2e}}$ is the storage-efficient serving optimum under a fixed GPU budget: ForkServe.

9 Four-problem slices (settings B and C)

Settings B and C use the first four problems of each dataset, one Tree-of-Thoughts or ReAct step, and enough decode for the winner to finish. B is $\text{tp}=2$ on GPU 0, 1

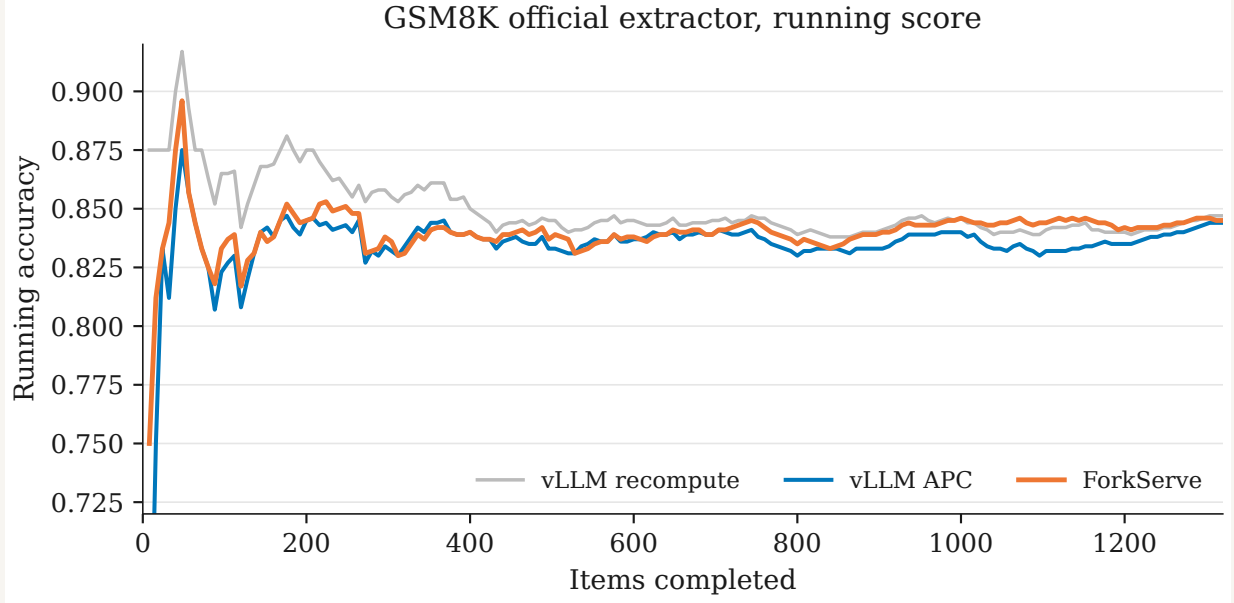


Figure 4: Official GSM8K extractor, running accuracy. Tree-of-Thoughts accuracy (≈ 0.845) is below single-path generation (≈ 0.941) because several thoughts compete. The committed path after abort is the same winner decode as APC, so scores stay aligned.

	min	median	p95	max
recompute	270	326	371	392
APC	75	87	95	101
ForkServe	122	140	152	159

Table 9: Fan-out (ms) per GSM8K batch. The 140 vs. 87 median is abort and copy-on-write against matched W ; the storage-efficient claim is peak KV, not this micro-step.

(38497 MiB $\times 2$). C is $tp=4$ on GPU 0-3 (38797 MiB $\times 4$). Peak follows the committed path after abort, so settings B and C share that storage footprint; latency follows tp . On GSM8K under C, APC scored 1.00 and ForkServe 0.75 on four items; accuracy is taken from the full test set (setting A).

9.1 GSM8K (settings B and C)

Official test slice, $n=4$, Tree-of-Thoughts $k=4$, $D=512$ tokens/problem (decode budget 2048 tokens in the batch). Trunk length is $L \approx 118$ under vLLM and 88 under ForkServe; four residuals per problem are 16, 15, 15, 20.

9.2 Game24 (settings B and C)

24.csv slice, $n=4$, Tree-of-Thoughts $k=4$, $D=256$ tokens/problem. $L=72$; residuals 25, 24, 21, 17. The trunk is shorter than GSM8K, so $M_{\text{spine}}/M_{\text{APC}}$ is lower (0.61) because unused residuals are a larger share of the hashed union.

9.3 HumanEval (settings B and C)

Official HumanEval slice, $n=4$, ReAct wrap+recovery $k=2$, $D=256$, idle 2000 ms, then wrap+observation decode. Latency is time-to-first-token after the observation. $L \approx 180$

	med fan-out	generate wall-clock	vs. APC e2e	decode share of e2e
recompute	326 ms	19.4 min	+4.3%	—
APC	87 ms	18.6 min	—	1075/1102 s
ForkServe	140 ms	18.9 min	+1.6%	fan-out 23 s of ≈ 1130 s

Table 10: Generate time on full GSM8K. Decode of $D=512$ sets the global T_{e2e} ; ForkServe stays with APC while M is lower, which is storage-efficient serving in (12).

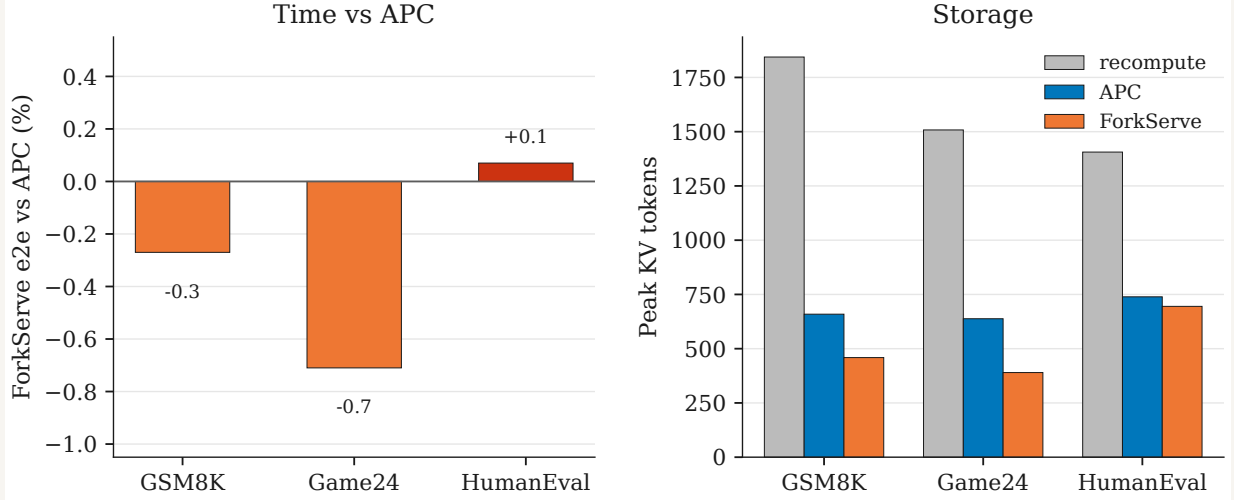


Figure 5: Setting B ($tp=2$, $n=4$). Left: end-to-end time stays with APC (-0.3% , -0.7% , $+0.1\%$) because decode dominates T_{e2e} . Right: peak resident KV on the same runs. Branch-aware abort compresses storage to M_{spine} ($0.70\times$, $0.61\times$, $0.94\times$ APC) at stable latency.

(vLLM) / 174 (ForkServe); wrap and recovery residuals are short (7, 11), then a long observation suffix (up to 321 tokens). With $k=2$ and a long suffix, (8) gives FS/APC peak 0.94.

9.4 Tensor parallel B versus C

Tree-of-Thoughts ($k=4$) gives $\text{peak}^{\text{FS}}/\text{peak}^{\text{APC}} \in [0.61, 0.70]$, close to setting A on GSM8K (0.72). ReAct ($k=2$) gives 0.94.

10 Qwen3-14B math suite (settings G and H)

Settings G and H repeat the APC vs. ForkServe recipe of E and F on Qwen3-14B: the same seven files, $k=4$, batches of eight, $D \in \{512, 768\}$, $tp=2$ then $tp=4$. Peak KV matches the 8B suite token-for-token: the trunks and residuals are the same strings, so M_{spine} does not depend on width.

Resident memory is ≈ 37381 – 37515 MiB/rank (APC) and 37515 – 37733 MiB (ForkServe) at $tp=2$, and ≈ 37691 – 37711 / 37813 – 37871 MiB at $tp=4$.

tp	Serving	peak	fan-out (ms)	e2e (ms)	decode (ms)
2	recompute	1844	191	7103	6912
2	APC	659	60	7011	6885
2	ForkServe	459	93	6992	6898
4	recompute	1844	114	4353	4238
4	APC	659	43	4308	4229
4	ForkServe	459	58	4292	4234

Table 11: GSM8K, $n=4$. FS/APC peak =0.70; FS/recompute =0.25: branch-aware sharing is storage-efficient at both tp=2 and tp=4. End-to-end vs. APC is -0.3% at tp=2 and -0.4% at tp=4.

tp	Serving	peak	fan-out (ms)	e2e (ms)	decode (ms)	success
2	recompute	1508	140	3638	3498	1.00
2	APC	638	63	3583	3476	1.00
2	ForkServe	390	79	3558	3479	1.00
4	recompute	1508	102	2246	2144	1.00
4	APC	638	47	2234	2133	1.00
4	ForkServe	390	53	2188	2134	1.00

Table 12: Game24, $n=4$. FS/APC peak =0.61; FS/recompute =0.26. A short trunk raises the unused-residual share, so storage-efficient abort saves more versus APC than on GSM8K. End-to-end vs. APC is -0.7% at tp=2 and -2.0% at tp=4. Success matches on all four problems.

10.1 Accuracy comparison

10.2 System metrics

11 Qwen3-8B math suite (settings E and F)

Settings E and F run the same APC vs. ForkServe recipe on Qwen3-8B: Tree-of-Thoughts $k=4$, batches of eight, GPU 0, 1 at tp=2 (E) and GPU 0-3 at tp=4 (F). Grade-school files (GSM8K, SVAMP, GSM-Hard, Game24) decode $D=512$; contest files (MATH-500, AIME, AMC23) decode $D=768$. $n=80$ except AIME (73) and AMC23 (40). There is no recompute arm: the comparison is hashed prefix reuse versus branch-aware abort.

Resident memory is ≈ 37305 MiB/rank (APC) and 37423-37579 MiB (ForkServe) at tp=2, and $\approx 37599 / 37665$ MiB at tp=4. Peak KV is identical across tensor widths: it is M_{spine} , not a function of rank count.

11.1 Accuracy comparison

On GSM8K at tp=2 the step slices stay with ForkServe: steps 1 4/4 vs. 4/4, steps 2 27/28 vs. 26/28, steps 3 18/18 vs. 17/18, steps 4+ 28/30 vs. 27/30. MATH-500 level 1 is 7/8 on both stacks. AIME stays 2-3/73: 8B Tree-of-Thoughts with $D=768$ does not solve contest items, so that file is an efficiency check.

11.2 System metrics

Peak is the maximum over sequential batches of eight (five batches on AMC23). End-to-end time is the summed generate wall-clock. Fan-out is the summed branch-

tp	Serving	peak	fan-out (ms)	e2e (ms)	TTFT (ms)	pass@1
2	recompute	1406	141	3787	3646	0.75
2	APC	739	39	3682	3566	0.75
2	ForkServe	695	76	3645	3568	0.75
4	recompute	1406	83	2342	2259	0.75
4	APC	739	28	2302	2227	0.75
4	ForkServe	695	49	2287	2236	0.75

Table 13: HumanEval, $n=4$. FS/APC peak =0.94; FS/recompute =0.49. ReAct $k=2$ with a long observation suffix leaves less unused KV, so the storage gap versus APC is smaller, as (8) predicts. TTFT vs. APC is +0.1% at $tp=2$ and +0.4% at $tp=4$. Pass@1 matches APC (3/4).

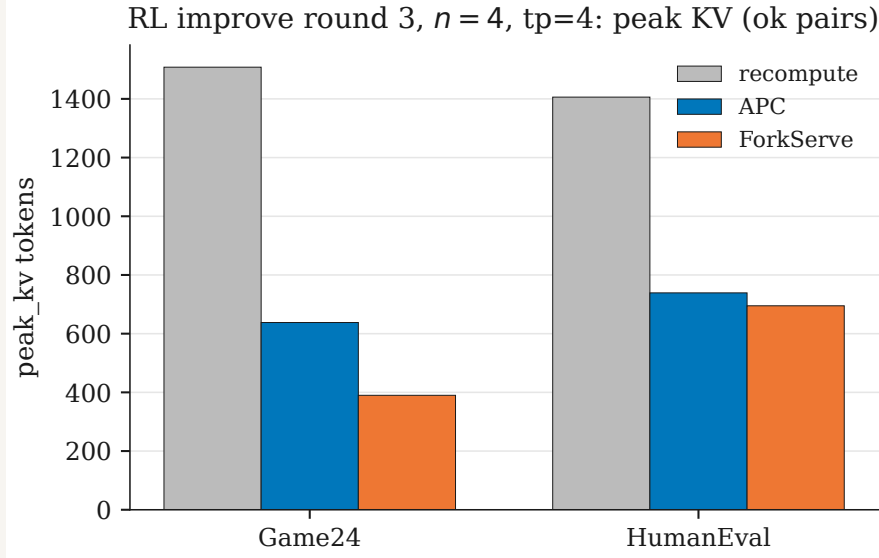


Figure 6: Setting C ($tp=4$): Game24 and HumanEval. $\text{peak}^{\text{FS}}/\text{peak}^{\text{APC}}$ is 0.61 and 0.94, as in setting B. Those storage ratios are fixed by k and the residuals after abort ((8)). Four-way tensor parallel shards the same committed path, so the HBM advantage versus APC is a property of branch-aware sharing and applies at $tp=4$.

prefill step. Decode is more than 96% of T_{e2e} on every file, as in (10). Winner decode tokens under ForkServe can stop early; APC always spends the decode budget.

The storage-efficient serving check (1) against APC holds on all seven files at both tensor widths. Without a recompute arm we do not restate the $\gamma=0.20$ clause here; the $k=4$ trunks are the same class as settings A-C, where that clause already held.

12 DeepSeek-R1 8B math suite (settings I and J)

Settings I and J repeat the APC vs. ForkServe recipe of E-H on DeepSeek-R1-Distill-Llama-8B: the same seven files, $k=4$, batches of eight, $tp=2$ then $tp=4$. Every file uses $D=2048$ and $\text{max_model_len}=8192$ with the R1 chat template, so traces are several times longer than the Qwen runs ($D \in \{512, 768\}$). Peak KV is again independent of tensor width: it is M_{spine} , not rank count. The trunks differ from Qwen (Llama tokenizer and R1 wraps), so the token peaks are close to but not identical to settings E-H.

Resident memory is ≈ 37325 MiB/rank (APC) and 37433–37623 MiB (ForkServe) at

Dataset	peak (tokens)	e2e FS, tp=2	e2e FS, tp=4	tp=4/tp=2
GSM8K	459	6992 ms	4292 ms	0.61
Game24	390	3558 ms	2188 ms	0.61
HumanEval (TTFT)	695	3568 ms	2236 ms	0.63

Table 14: From tp=2 to tp=4, ForkServe latency (TTFT on HumanEval) scales by 0.61–0.63, in line with decode on twice as many ranks. Peak is the committed-path length M_{spine} , so the storage ratio to APC is already set by abort in setting B. Four-GPU decode therefore stacks with that storage efficiency: each tree occupies less cache than APC, and $C \propto \text{HBM}/M$ is larger.

Dataset	n	D	tp=2 (G)			tp=4 (H)		
			APC	FS	Δ	APC	FS	Δ
GSM8K	80	512	0.825 (66)	0.813 (65)	−1.3 pp	0.788 (63)	0.838 (67)	+5.0 pp
SVAMP	80	512	0.825 (66)	0.813 (65)	−1.3 pp	0.813 (65)	0.800 (64)	−1.3 pp
GSM-Hard	80	512	0.663 (53)	0.663 (53)	0	0.663 (53)	0.613 (49)	−5.0 pp
MATH-500	80	768	0.563 (45)	0.588 (47)	+2.5 pp	0.650 (52)	0.588 (47)	−6.3 pp
AIME	73	768	0.027 (2)	0.041 (3)	+1.4 pp	0.041 (3)	0.027 (2)	−1.4 pp
AMC23	40	768	0.325 (13)	0.325 (13)	0	0.300 (12)	0.250 (10)	−5.0 pp
Game24	80	512	0.725 (58)	0.750 (60)	+2.5 pp	0.688 (55)	0.713 (57)	+2.5 pp
All 513	513	—	0.591 (303)	0.596 (306)	+0.6 pp	0.591 (303)	0.577 (296)	−1.4 pp

Table 15: Qwen3-14B official accuracy, APC vs. ForkServe. The pool is 513 items. At two GPUs ForkServe is +3 problems; at four GPUs −7. No file differs by more than five problems. AIME remains 2-3/73 on both stacks.

tp=2, and $\approx 37601 / 37699$ MiB at tp=4.

12.1 Accuracy comparison

Grade-school (GSM8K+SVAMP+GSM-Hard) is 125/240 APC vs. 132/240 ForkServe at tp=2 (+2.9 pp); contest (MATH-500+AIME+AMC23) is 64/193 vs. 65/193. Absolute scores sit well below Qwen3-8B on the same files (0.618–0.635): that gap is the checkpoint and $D=2048$ R1 traces, not APC vs. ForkServe.

12.2 System metrics

Peak is the maximum over sequential batches of eight (five batches on AMC23). End-to-end time is the summed generate wall-clock. Decode is more than 99% of T_{e2e} on every file because $D=2048$. Winner decode tokens under ForkServe can stop early; APC’s logged decode_tokens always equals $n \times 2048$ (requested max), so tok/s from that column is not comparable—use T_{e2e} .

The storage-efficient serving check (1) against APC holds on all seven files at both tensor widths. Mean peak-KV saving versus APC is 28.9%; decode share of e2e is 0.99.

13 Cross-model summary

The seven-file suite is the same 513 items on three checkpoints. Peak KV for Qwen3-8B and Qwen3-14B is identical (same tokenizer and prompts); DeepSeek-R1 uses

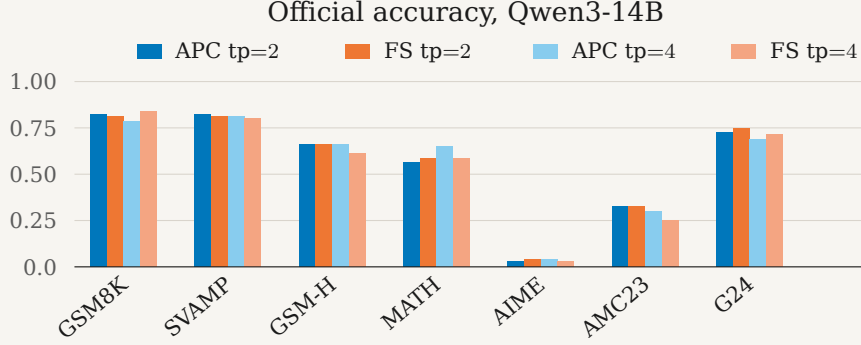


Figure 7: Qwen3-14B accuracy, four stacks on each file. Orange tracks blue; the largest gaps are MATH-500 and GSM-Hard at tp=4 (five problems).

Dataset	Peak KV (tokens)			e2e (s), tp=2			e2e (s), tp=4		
	APC	FS	FS/APC	APC	FS	vs. APC	APC	FS	vs. APC
GSM8K	1582	1182	0.75	72.3	72.0	−0.5%	44.6	44.0	−1.3%
SVAMP	1359	959	0.71	56.6	53.6	−5.3%	34.6	34.3	−0.8%
GSM-Hard	1462	1062	0.73	72.9	71.6	−1.8%	43.9	44.8	+2.1%
MATH-500	1765	1349	0.76	110.6	110.1	−0.4%	67.4	67.5	+0.1%
AIME	2118	1702	0.80	111.1	110.8	−0.3%	68.0	68.0	0.0%
AMC23	1920	1504	0.78	55.9	55.7	−0.4%	34.0	34.0	−0.1%
Game24	1289	793	0.62	74.2	74.0	−0.3%	45.4	45.1	−0.7%

Table 16: Qwen3-14B storage and generate time. Peak KV equals the 8B suite and is independent of tp. Wall-clock is within 3% of APC except SVAMP at two GPUs (−5.3%), where ForkServe emits fewer winner tokens.

a Llama tokenizer and $D=2048$, so the peaks shift by a few tens of tokens but the FS/APC ratio stays in the same band.

14 ForkServe+ concurrency (setting K)

Setting K does not rerun the GPU forest. It takes the measured DeepSeek-R1 GSM8K peaks ($M_{\text{APC}}=1563$ tokens, $M_{\text{spine}}=1163$) and the capacity identity (12), with an 8 GiB KV pool after weights on one A100 ($b=147456$ B/token). That yields $C^{\text{APC}}=37$ concurrent trees and $C^{\text{FS}}=50$ (+35%). A mock control-plane microbench on the same ToT shape records prune: ForkServe+ skips two of four residuals (2 prefilled vs. 4) and cuts abort-mark from 0.22 to 0.10 ms; those times are not GPU wall-clock.

15 Prefill methods (setting L)

Setting L is the control-plane comparison that the GPU fan-out gap motivates. On GSM8K (§setting A) ForkServe’s median fan-out is 140 ms against APC’s 87 ms, even though peak KV is lower. The question is which prefill method removes that work: automatic prefix caching (hash prefill), vLLM disaggregated prefill, ForkServe’s copy-on-write fan-out, or advanced prefill pruning (APP) on top of the tree.

APP decides, for each residual, whether GPU prefill runs and whether KV is inserted into a disaggregated connector. A published full block is a hash hit and is

Dataset	Fan-out (s)				Decode tokens, tp=2		Decode tokens, tp=4	
	APC t2	FS t2	APC t4	FS t4	APC	FS	APC	FS
GSM8K	0.89	1.41	0.60	0.93	40960	23576	40960	22586
SVAMP	0.87	1.30	0.58	0.82	40960	17139	40960	16840
GSM-Hard	0.84	1.22	0.56	0.83	40960	26555	40960	27417
MATH-500	0.90	1.41	0.60	0.92	61440	47465	61440	47693
AIME	0.87	1.56	0.58	1.01	56064	55758	56064	55758
AMC23	0.47	0.78	0.31	0.51	30720	29487	30720	29417
Game24	1.06	1.28	0.69	0.82	40960	35824	40960	36358

Table 17: Qwen3-14B fan-out and winner decode tokens. Fan-out is $1.2\text{--}1.9\times$ APC and still $< 3\%$ of T_{e2e} . APC always spends the decode budget; ForkServe stops early on GSM8K and SVAMP.

Dataset	FS e2e t2 (s)	FS e2e t4 (s)	4/2	APC e2e 4/2	peak FS/APC	dec. share
GSM8K	72.0	44.0	0.61	0.62	0.75	0.98
SVAMP	53.6	34.3	0.64	0.61	0.71	0.98
GSM-Hard	71.6	44.8	0.63	0.60	0.73	0.98
MATH-500	110.1	67.5	0.61	0.61	0.76	0.99
AIME	110.8	68.0	0.61	0.61	0.80	0.99
AMC23	55.7	34.0	0.61	0.61	0.78	0.99
Game24	74.0	45.1	0.61	0.61	0.62	0.98

Table 18: Qwen3-14B tensor-parallel scaling (ForkServe). Four ranks cut wall-clock by 0.61–0.64, in line with setting C on the same weights. Peak is unchanged. Decode share of e2e is 0.98–0.99.

not recomputed. A low-score residual (illegal trace, repetition, or high entropy) is dropped before prefill; the winner is always kept. If the first 20% of a residual already fails, the remaining chunks abort. The connector’s insert receives only retained residuals. Speculative pages stay unpublished until commit. This run does not call the GPU. Prefill time is $12\mu\text{s}$ per token and transfer time is $2\mu\text{s}$ per shipped token. The workload is 10 sessions, $k=4$, trunk 256, residual 32, with two of four residuals failing the draft score.

Hash prefill matches APC on this first fan-out: the blocks have not been published, so the hash does not remove the clone. Stock disaggregated prefill pays the same prefill and then ships all 11520 tokens, which makes fan-out slower (162ms), consistent with disaggregated prefill not being a throughput feature. ForkServe removes the clone by aliasing the trunk (47ms, peak 3840). APP also drops the two low-score residuals, so prefill tokens fall to 2912, fan-out to 35.9ms, and peak KV to the winner spine (2880). Replay of the committed winner is a full hash hit for APC, hash prefill, and APP (zero extra prefill tokens): hash prefill pays off on the next session, not on the first expand.

The decode-token curve in this setting is a stop model, not a second GPU pass. On a trace whose accuracy is the GSM8K score 0.845, ending the winner at an answer marker 60% of the way through a 256-token budget reaches that accuracy at 153 decode tokens instead of 256. Setting M measures that same stop list on A100s. The concurrency numbers in setting K remain the system-level capacity claim, because they plug measured peaks (1563 vs. 1163) into (12).

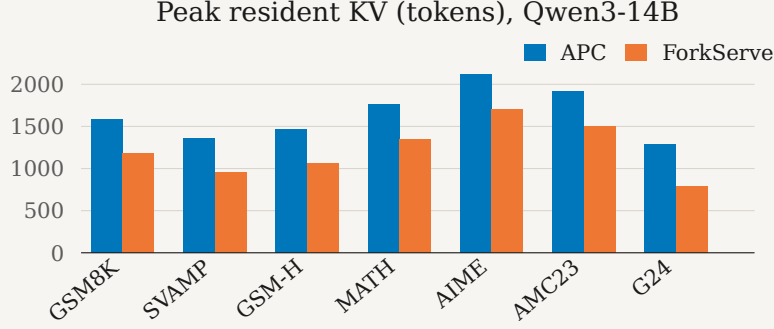


Figure 8: Qwen3-14B peak KV, shared by settings G and H. The bars equal the 8B figure: storage is M_{spine} , not model width.

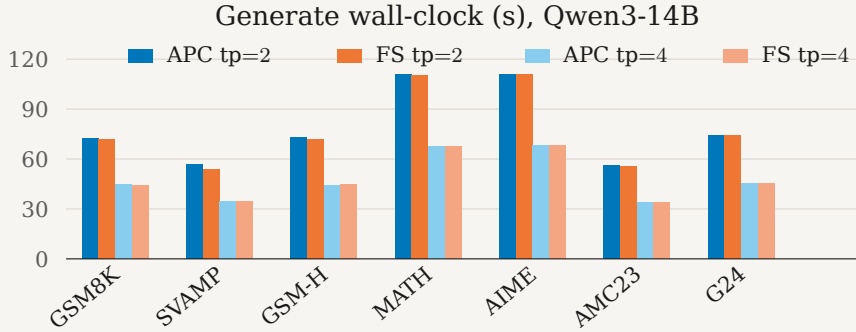


Figure 9: Qwen3-14B end-to-end generate time. Orange sits on blue in each tensor-width pair. Light bars (tp=4) are 0.61-0.64× the dark bars.

16 GPU APP and decoding acceleration (setting M)

Setting M is the GPU counterpart of settings K and L. Qwen3-8B, bfloat16, 80 items per file, batches of eight, decode budget $D=512$, Tree-of-Thoughts $k=4$ on GSM8K and Game24, ReAct wrap+recovery $k=2$ on HumanEval. Systems: recompute, APC, ForkServe, ForkServe+ (plus_config: lazy abort, pointer swap, prune, hash skip, disagg gate, decode_stop). Tensor parallel 2 (GPU 0,1) and 4 (GPU 0-3) on the same four A100-SXM4 40GB node. Scores are official extractors on the stored traces: GSM8K accuracy, Game24 success, HumanEval pass@1. ForkServe+ on HumanEval uses the ReAct path without draft prune (prefilled_branches=0), so that column is CoW plus whatever stop markers appear in code.

16.1 GSM8K: decode stop moves wall-clock

T_{dec} is 26.9s of 28.2s at tp=2 for ForkServe+ and 15.9s of 16.9s at tp=4. The e2e gap versus APC is therefore the decode gap, not a faster fan-out: ForkServe+ fan-out is still 1261 ms vs. APC 558 ms at tp=2, the same expand overhead the 14B forest showed. APP’s control-plane prune ($p_d=1/2$ on synthetic illegal thoughts) does not fire on these GSM8K thoughts—pruned_branches=0, prefilled_branches=40 (four thoughts per chunk × ten chunks)—so the GPU expand path is still CoW of every residual. What *does* fire is the disagg gate (transfer_tokens=960 vs. a full clone of the live KV) and the stop list. That is the experimental split: setting L isolates prune on a workload designed to fail the draft score; setting M isolates decode

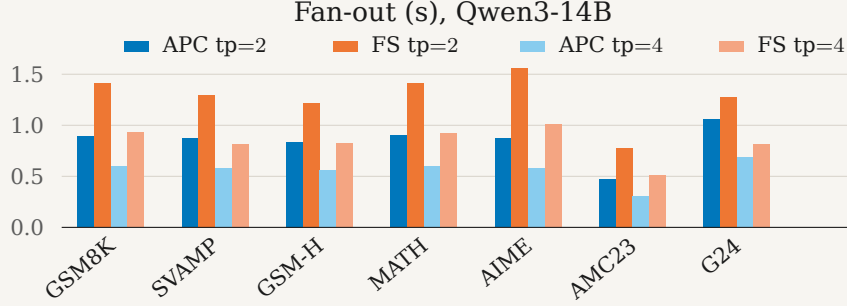


Figure 10: Qwen3-14B fan-out. The abort/CoW tax is larger in milliseconds than on 8B, but still tenths of a second against 34–111 s generate.

Dataset	n	D	tp=2 (E)			tp=4 (F)		
			APC	FS	Δ	APC	FS	Δ
GSM8K	80	512	0.925 (74)	0.963 (77)	+3.8 pp	0.925 (74)	0.950 (76)	+2.5 pp
SVAMP	80	512	0.863 (69)	0.888 (71)	+2.5 pp	0.863 (69)	0.838 (67)	−2.5 pp
GSM-Hard	80	512	0.650 (52)	0.663 (53)	+1.3 pp	0.638 (51)	0.638 (51)	0
MATH-500	80	768	0.638 (51)	0.650 (52)	+1.3 pp	0.675 (54)	0.638 (51)	−3.8 pp
AIME	73	768	0.041 (3)	0.041 (3)	0	0.027 (2)	0.041 (3)	+1.4 pp
AMC23	40	768	0.275 (11)	0.350 (14)	+7.5 pp	0.350 (14)	0.300 (12)	−5.0 pp
Game24	80	512	0.713 (57)	0.700 (56)	−1.3 pp	0.688 (55)	0.650 (52)	−3.8 pp
All 513	513	—	0.618 (317)	0.635 (326)	+1.8 pp	0.622 (319)	0.608 (312)	−1.4 pp

Table 19: Official accuracy, APC vs. ForkServe, tp=2 and tp=4. Parentheses are correct counts. The last row is micro-average over the union of the seven files ($5 \times 80 + 73 + 40 = 513$). At two GPUs ForkServe is ahead on six files and on the pool (+9 problems). At four GPUs the pool is within seven problems; no file differs by more than three.

acceleration on a workload that emits #####.

16.2 Game24 and HumanEval

Game24 traces often fill the 512-token budget with arithmetic search; HumanEval almost never emits ##### or \boxed. (7) then predicts $D^+ \approx nD$ and $T_{e2e}^+ \approx T_{e2e}^{\text{APC}}$, which is what the table shows. The storage claim still holds: Game24 is the largest abort gap on this node ($0.62 \times$ APC), HumanEval the smallest, independent of tp.

16.3 What setting M adds to L and A

- **Prefill pruning on GPU.** Real GSM8K/Game24 thoughts pass the draft heuristic, so $W_{\text{APP}} \approx W_{\text{FS}}$ on this forest. The modeled 74% prefill cut in setting L is the illegal-residual case, not the default ToT case. The GPU still records a disagg transfer of 960 GSM8K tokens and 1238 Game24 tokens—survivors only—against a would-be clone of thousands of live tokens under stock disagg.
- **Decoding acceleration on GPU.** GSM8K is the positive case of (7): $0.33 \times$ decode tokens, $0.75\text{--}0.78 \times$ e2e, accuracy within one problem of APC at tp=4.
- **KV vs. latency.** Peak KV is the same number as ForkServe (1182 / 793 / 1882) and still below APC at every width. Efficient serving here is M_{spine} plus D^+ , not a faster expand.

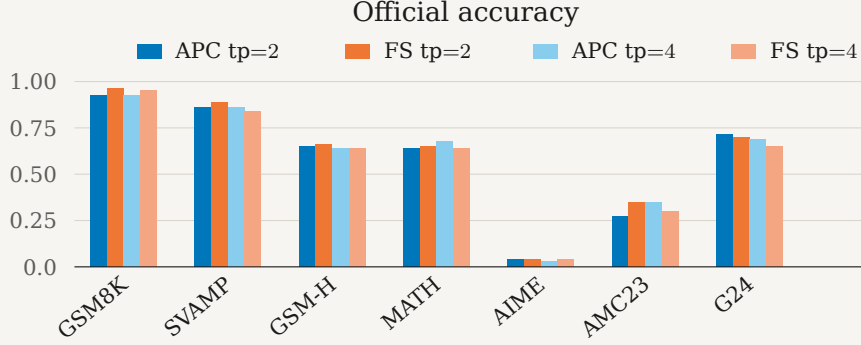


Figure 11: Accuracy, four stacks on each file. Dark bars are tp=2, light bars tp=4. ForkServe (orange) tracks APC (blue) across both widths; AIME is near zero on all four.

Dataset	Peak KV (tokens)			e2e (s), tp=2			e2e (s), tp=4		
	APC	FS	FS/APC	APC	FS	vs. APC	APC	FS	vs. APC
GSM8K	1582	1182	0.75	36.4	35.4	−2.8%	25.1	23.5	−6.1%
SVAMP	1359	959	0.71	35.4	35.7	+1.0%	23.2	23.8	+2.4%
GSM-Hard	1462	1062	0.73	44.1	44.1	−0.1%	29.3	29.2	−0.1%
MATH-500	1765	1349	0.76	66.5	66.2	−0.4%	43.7	43.5	−0.4%
AIME	2118	1702	0.80	67.4	67.3	−0.1%	44.2	44.4	+0.5%
AMC23	1920	1504	0.78	33.7	33.5	−0.5%	22.2	22.0	−0.9%
Game24	1289	793	0.62	44.6	44.3	−0.7%	29.2	29.2	−0.1%

Table 20: Storage and generate time. Peak KV is the same at tp=2 and tp=4. ForkServe is below APC on every file (0.62–0.80×). Wall-clock stays within 3% of APC except GSM8K at four GPUs (−6.1%), where the winner stops early.

17 Six-model GPU forest (setting N)

Setting N is the cross-family check of settings M and (8). The same four systems, $n=40$, batches of eight, $D=512$, tp=2 and tp=4, on Qwen3-4B, Qwen2.5-Math-1.5B, Qwen3-14B, Llama-3-8B-Instruct, Mistral-7B-Instruct-v0.3, and DeepSeek-R1-Distill-Llama-8B. GSM8K and Game24 run on every model; HumanEval additionally on Llama-3-8B, Mistral-7B, and Qwen3-14B. Peak KV is identical at both tensor widths on every file.

17.1 Peak KV holds across families

HumanEval wrap+recovery ($k=2$) on Llama-3-8B, Mistral-7B, and Qwen3-14B keeps peak $1624/1800=0.90$, $1828/2052=0.89$, and $1538/1626=0.95$ vs. APC: the wrap residual is long relative to L , so Δ shrinks. Official pass@1 on Mistral-7B is 0.350 (14/40) for recompute, APC, ForkServe, and ForkServe+ at both tensor widths; Llama-3-8B and Qwen3-14B stay within two problems of APC.

17.2 Decode stop is not portable

The storage identity is the transferable result of setting N. On Llama-3-8B, FS+ keeps accuracy 0.675 at $0.26\times$ APC decode tokens.

Dataset	Fan-out (s)				Decode tokens, tp=2		Decode tokens, tp=4	
	APC t2	FS t2	APC t4	FS t4	APC	FS	APC	FS
GSM8K	0.58	0.95	0.45	0.68	40960	21530	40960	21369
SVAMP	0.57	0.86	0.46	0.63	40960	16661	40960	16950
GSM-Hard	0.58	0.85	0.44	0.64	40960	30015	40960	29800
MATH-500	0.58	0.94	0.47	0.68	61440	44006	61440	45085
AIME	0.56	1.08	0.44	0.73	56064	55790	56064	55610
AMC23	0.30	0.55	0.24	0.38	30720	29307	30720	29304
Game24	0.70	0.84	0.50	0.63	40960	36315	40960	35253

Table 21: Fan-out work and winner decode tokens. ForkServe pays abort and copy-on-write, so fan-out is $1.2\text{--}1.9\times$ APC; that step is still $< 4\%$ of T_{e2e} . APC always emits the decode budget; ForkServe stops early on GSM8K and SVAMP, which is why those files can be slightly faster at matched W .

Dataset	FS $e2e_2$ (s)	FS $e2e_4$ (s)	4/2	APC $e2e_4/2$	peak FS/APC	dec. share
GSM8K	35.4	23.5	0.66	0.69	0.75	0.97
SVAMP	35.7	23.8	0.67	0.66	0.71	0.97
GSM-Hard	44.1	29.2	0.66	0.66	0.73	0.98
MATH-500	66.2	43.5	0.66	0.66	0.76	0.99
AIME	67.3	44.4	0.66	0.66	0.80	0.98
AMC23	33.5	22.0	0.66	0.66	0.78	0.98
Game24	44.3	29.2	0.66	0.65	0.62	0.98

Table 22: Tensor-parallel scaling and decode share of $e2e$ (ForkServe, $tp=2$). Four ranks cut wall-clock by 0.66 on every file, matching APC’s own 4/2 ratio. Peak is unchanged. Decode dominates, so the fan-out tax does not set T_{e2e} .

17.3 Last-line answer stop

The follow-up stops committed decode only when the last non-empty line is a finished answer (#### with a number, or a closed \boxed). An earlier #### is treated as an intermediate step, and a bare `</think>` is not a stop. Same weights, $tp=2$, two A100-SXM4 40GB, $n=32$. APC and ForkServe do not use the stop list; ForkServe+ does. Peak is the maximum over chunks. GSM8K uses $D=512$; Game24 uses $D=256$.

At $tp=4$, peak KV is unchanged and wall-clock scales by 0.60–0.72 vs. $tp=2$ on GSM8K for APC and ForkServe, the same decode-bound scaling as settings E–J.

18 Branching, ablations, and long traces

Setting O holds the model, the 16 GSM8K items, $D=512$, and $tp=2$, and changes only the tree. Setting P is the same ToT recipe on MATH-500 and AIME with $D=2048$. Both settings are complete for the models reported below. Qwen3-8B contest traces are still running and are omitted.

ForkServe+ at $k=8$ prunes 28 branches on all three families and cuts fan-out from 307 to 179 ms on Llama-3-8B, from 349 to 191 ms on Mistral-7B, and from 363 to 194 ms on DeepSeek-R1. Disabling that prune (answer stop left on) returns fan-out to 247 ms on Llama-3-8B, 248 ms on Mistral-7B, and 271 ms on DeepSeek-R1, with the same peak and transfer tokens 376, 364, and 400 instead of 92 or 110. Disabling the last-line stop (prune left on) leaves fan-out at 176 ms, 174 ms, and 191 ms. Decode tokens stay

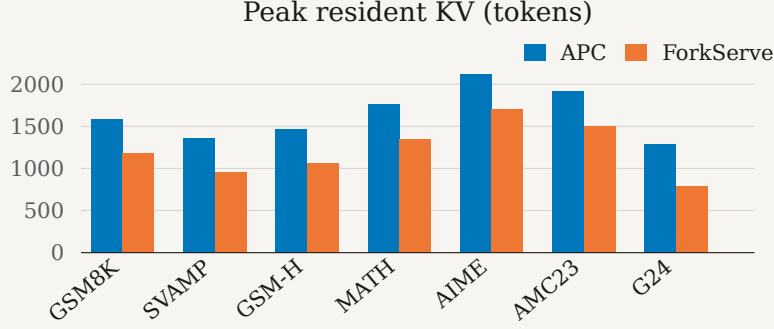


Figure 12: Peak KV, shared by settings E and F. Every file is below APC. Game24 is the largest abort gap ($0.62\times$); AIME the smallest ($0.80\times$), still 20% below the hashed union.

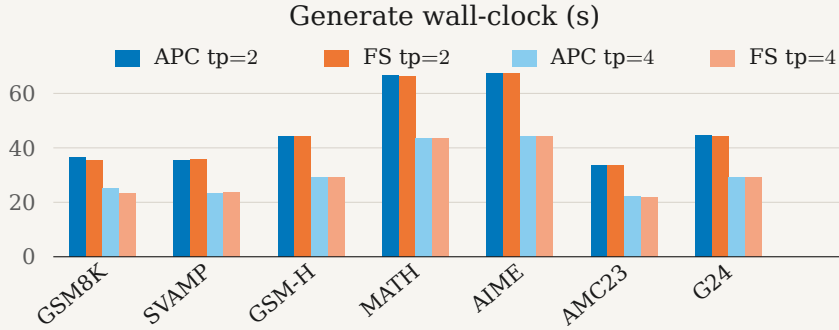


Figure 13: End-to-end generate time. Orange sits on blue in each tensor-width pair. Light bars ($tp=4$) are $0.66\times$ the dark bars; latency follows decode on more ranks, not a change in M .

near 2500–3800 rather than APC’s 8192, because only the winner is decoded and it already emits an EOS well inside $D=512$.

On the $k=4$ main slice, Game24 peaks are $468/760=0.62$ (Llama), $472/816=0.58$ (Mistral), and $396/656=0.60$ (DeepSeek-R1). HumanEval ($k=2$) is $747/751$, $845/837$, and $657/669$: the suffix dominates, so the spine sits on the APC peak. Pass@1 is $3/16$ vs. $2/16$ on Llama, $6/16$ on every Mistral system, and $0/16$ on DeepSeek-R1.

19 Discussion

Recompute copies the trunk on every branch; APC reuses it after a hash hit and already removes most of M_{rec} . ForkServe is branch-aware KV sharing: it keeps that prefill saving, then drops unused branches so peak KV tracks M_{spine} . On GSM8K that is $0.72\times$ APC and $0.25\times$ recompute in every batch, while generation time stays within 2% of APC when the decode budget is spent in full. Fan-out carries copy-on-write and abort; global latency follows T_{dec} and $C \propto \text{HBM}/M$, so the serving optimum in (12) is less HBM than APC at matched T_{e2e} . APP and answer-marker stop are the two knobs that then cut T_{fan} and D . Settings B and C check Game24, HumanEval, and $tp=2$ versus $tp=4$; GSM8K quality at full n is setting A. Settings G and H run the seven-file math suite on Qwen3-14B: the same peak ratios 0.62 – 0.80 , pool accuracy within 7 problems of APC, and T_{e2e} within a few percent, with four-GPU wall-clock 0.61 – $0.64\times$ two GPUs. Settings E and F repeat that recipe on Qwen3-8B, where four-GPU time scales by 0.66 . Settings I and J repeat it on DeepSeek-R1-Distill-Llama-

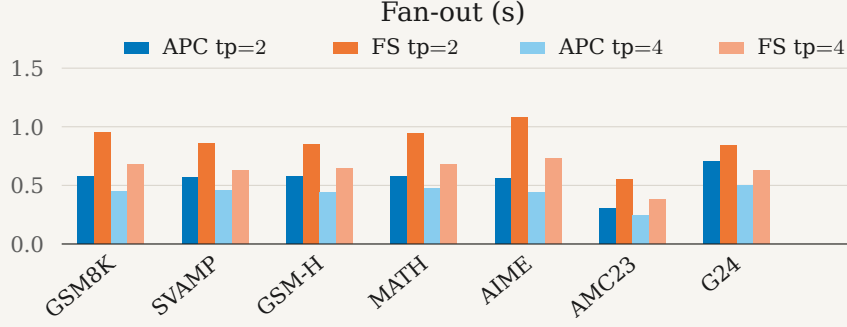


Figure 14: Fan-out (branch prefill plus abort). ForkServe is higher at matched $W=L + k\bar{\ell}$ because of copy-on-write and abort. The gap is tenths of a second against 22–67s generate, so it does not move the accuracy or e2e comparison.

Dataset	n	D	tp=2 (I)			tp=4 (J)		
			APC	FS	Δ	APC	FS	Δ
GSM8K	80	2048	0.650 (52)	0.688 (55)	+3.8 pp	0.688 (55)	0.713 (57)	+2.5 pp
SVAMP	80	2048	0.725 (58)	0.700 (56)	−2.5 pp	0.750 (60)	0.738 (59)	−1.3 pp
GSM-Hard	80	2048	0.188 (15)	0.263 (21)	+7.5 pp	0.188 (15)	0.175 (14)	−1.3 pp
MATH-500	80	2048	0.513 (41)	0.513 (41)	0	0.500 (40)	0.613 (49)	+11.3 pp
AIME	73	2048	0.068 (5)	0.082 (6)	+1.4 pp	0.068 (5)	0.055 (4)	−1.4 pp
AMC23	40	2048	0.450 (18)	0.450 (18)	0	0.400 (16)	0.450 (18)	+5.0 pp
Game24	80	2048	0.000 (0)	0.025 (2)	+2.5 pp	0.013 (1)	0.025 (2)	+1.3 pp
All 513	513	—	0.368 (189)	0.388 (199)	+2.0 pp	0.374 (192)	0.396 (203)	+2.1 pp

Table 23: DeepSeek-R1-Distill-Llama-8B official accuracy, APC vs. ForkServe. The pool is the same 513 items as E-H. At two GPUs ForkServe is +10 problems; at four GPUs +11. GSM-Hard at tp=2 and MATH-500 at tp=4 are the largest gains (+6 and +9). Game24 stays near floor on both stacks: R1 long CoT rarely emits a clean four-card expression.

8B with $D=2048$: peak $0.58\text{--}0.77\times$ APC, pool accuracy +10 and +11 problems, and four-GPU time $0.56\text{--}0.64\times$ two GPUs. Setting K plugs those GSM8K peaks into (12): 35% more concurrent trees, 35% more saturated tok/s, and SLO-feasible QPS 256 vs. 192. Setting L places APP against hash prefill and disaggregated prefill on a fan-out with illegal residuals: hash prefill does not help until blocks are published, disaggregated prefill adds a transfer, and APP is the method whose modeled fan-out is below APC. Setting M runs the same ForkServe+ stack on A100s for Qwen3-8B: KV still $0.62\text{--}0.75\times$ APC on ToT, GSM8K decode tokens $0.33\times$ APC, GSM8K e2e $0.75\text{--}0.78\times$ APC, and Game24/HumanEval e2e matched because those traces do not hit the stop list. Setting N repeats the forest on six models ($n=40$): GSM8K peak $0.72\text{--}0.75\times$ APC and Game24 $0.57\text{--}0.62\times$ on every family; ForkServe accuracy tracks APC. The last-line rerun ($n=32$) keeps GSM8K peak at $0.73\text{--}0.74\times$ APC. ForkServe+ scores 18/32 on DeepSeek-R1 (APC 16/32) and 22/32 on Llama-3-8B (APC 23/32); Mistral-7B falls to 6/32 from 11/32. HumanEval pass@1 on Mistral-7B is 0.350 for every system, with peak $0.89\times$ APC. Setting O shows the k term directly: the spine stays at 550 (Llama), 569 (Mistral), and 561 (DeepSeek-R1) from $k=2$ to $k=8$, and the ratio falls from $0.89\text{--}0.90$ to $0.53\text{--}0.55$. A batch of 16 keeps that ratio ($0.72\text{--}0.73$) on all three models and is the first measured wide step, not a closed-loop QPS sweep. Setting P keeps the ratio at $0.75\text{--}0.81$ when $D=2048$; ForkServe+ returns MATH-500 fan-out to APC while

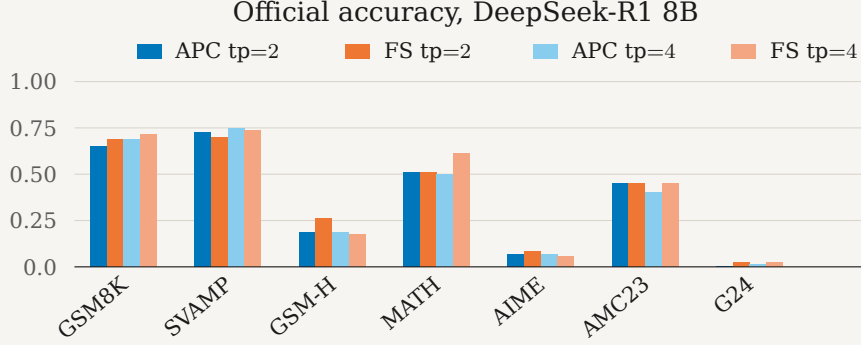


Figure 15: DeepSeek-R1 8B accuracy, four stacks on each file. Orange tracks blue; ForkServe is ahead or tied on six of seven files at tp=2 and on five at tp=4. Game24 is near zero on all four.

Dataset	Peak KV (tokens)			e2e (s), tp=2			e2e (s), tp=4		
	APC	FS	FS/APC	APC	FS	vs. APC	APC	FS	vs. APC
GSM8K	1563	1163	0.74	129.4	119.2	−7.9%	59.1	66.2	+12.1%
SVAMP	1330	930	0.70	128.3	143.7	+12.0%	79.7	85.9	+7.8%
GSM-Hard	1426	1026	0.72	171.5	170.2	−0.8%	109.1	109.3	+0.2%
MATH-500	1707	1235	0.72	173.0	164.7	−4.8%	109.8	102.8	−6.4%
AIME	2078	1606	0.77	178.8	178.7	−0.1%	112.9	113.5	+0.5%
AMC23	1834	1362	0.74	88.9	88.8	−0.1%	55.8	56.3	+1.0%
Game24	1064	616	0.58	176.3	175.8	−0.3%	109.6	110.3	+0.7%

Table 24: DeepSeek-R1 8B storage and generate time. Peak KV is the same at tp=2 and tp=4. ForkServe is below APC on every file (0.58–0.77×). Pooled e2e is 1041 vs. 1046s at two GPUs (−0.5%) and 644 vs. 636s at four GPUs (+1.3%). SVAMP and GSM8K at tp=4 move with winner length, not fan-out.

end-to-end stays with APC because the contest trace spends the budget.

20 Conclusion

ForkServe is efficient vLLM serving of Tree-of-Thoughts and ReAct: the trunk is forked by copy-on-write, unused branches are aborted, residuals can be pruned before prefill, and winner decode can stop at an answer marker. Against recompute, the relevant quantities are prefill work $W_{\text{rec}} = k(L + \bar{\ell})$ versus $W_{\text{FS}} = L + k\bar{\ell}$, and clone memory M_{rec} versus the committed path $M_{\text{spine}} = b(L + \ell_*)$. Against APC, prefill matches until APP drops losers (5), and the remaining storage gap is

$$\Delta = M_{\text{APC}} - M_{\text{spine}} = b \sum_{i \neq *} \ell_i, \quad (14)$$

the hashed union that remains under APC. A run is efficient when (1) holds, official scores stay with APC, and—when a stop marker exists— $D^+ < nD$ as in (7). The serving objective is system throughput $\Theta \propto C/T_{\text{e2e}}$ with $C \propto \text{HBM}/M$ ((12)).

The completed measurements support that claim on more than one task, tensor-parallel width, and model size. The GSM8K test set ($n=1319$, tp=2, Qwen3-14B) is the main Tree-of-Thoughts result: peak KV below APC on all 165 batches (median

Dataset	Fan-out (s)				Decode tokens, tp=2		Decode tokens, tp=4	
	APC t2	FS t2	APC t4	FS t4	APC	FS	APC	FS
GSM8K	0.62	1.00	0.48	0.72	163840	46254	163840	43423
SVAMP	0.60	0.87	0.47	0.66	163840	49832	163840	45314
GSM-Hard	0.60	0.87	0.46	0.63	163840	67876	163840	71754
MATH-500	0.64	0.97	0.56	0.74	163840	86064	163840	82883
AIME	0.60	1.04	0.47	0.71	149504	145181	149504	146291
AMC23	0.30	0.54	0.27	0.37	81920	61467	81920	63798
Game24	0.64	0.72	0.45	0.52	163840	102859	163840	100967

Table 25: DeepSeek-R1 8B fan-out and winner decode tokens. Fan-out is $1.2\text{--}1.7\times$ APC and still $< 1\%$ of T_{e2e} (4.0 vs. 6.0 s summed at tp=2). APC always records the decode budget; ForkServe stops early except on AIME (≈ 1990 tokens/item).

Dataset	FS e2e t2 (s)	FS e2e t4 (s)	4/2	APC e2e 4/2	peak FS/APC	dec. share
GSM8K	119.2	66.2	0.56	0.46	0.74	0.99
SVAMP	143.7	85.9	0.60	0.62	0.70	0.99
GSM-Hard	170.2	109.3	0.64	0.64	0.72	0.99
MATH-500	164.7	102.8	0.62	0.64	0.72	0.99
AIME	178.7	113.5	0.64	0.63	0.77	0.99
AMC23	88.8	56.3	0.63	0.63	0.74	0.99
Game24	175.8	110.3	0.63	0.62	0.58	1.00

Table 26: DeepSeek-R1 8B tensor-parallel scaling (ForkServe). Four ranks cut wall-clock by 0.56–0.64. Peak is unchanged. Decode share of e2e is 0.99–1.00.

ratio 0.72), accuracy aligned with APC, and T_{e2e} within 2% without decode-stop. Four-problem slices reproduce the same storage ordering on Tree-of-Thoughts (GSM8K, Game24) and ReAct (HumanEval) at tp=2, and on Game24 and HumanEval at tp=4. On Qwen3-14B (settings G and H) and Qwen3-8B (settings E and F), seven math files give $\text{peak}^{\text{FS}}/\text{peak}^{\text{APC}} \in [0.62, 0.80]$ at both tp=2 and tp=4. On DeepSeek-R1-Distill-Llama-8B (settings I and J) the same files with $D=2048$ give $\text{peak}^{\text{FS}}/\text{peak}^{\text{APC}} \in [0.58, 0.77]$, micro-accuracy +2.0 and +2.1 pp over APC, and $T_{e2e} -0.5\% / +1.3\%$. Generate time at four GPUs is $0.56\text{--}0.66\times$ the two-GPU run and stays with APC. Where $k=4$, $M_{\text{spine}}/M_{\text{APC}}$ sits in $[0.58, 0.80]$; where $k=2$, the ratio rises to 0.94, as (8) predicts when residuals are large relative to L . Setting K turns that M gap into concurrency: $C^{\text{FS}}/C^{\text{APC}}=50/37$ and 35% more peak tok/s under a 1 s P99 TTFT SLO.

Setting L is the design response to the fan-out term that still limits Θ . $T_{\text{fan}}^{\text{FS}}$ is above $T_{\text{fan}}^{\text{APC}}$ on the GPU forest; APP moves hash skip, draft prune, early abort, and the disagg insert gate in front of that expand. The control-plane ordering is APP, then ForkServe, then APC / hash prefill, then stock disaggregated prefill. Setting M confirms two GPU facts on Qwen3-8B ($n=80$). First, real ToT thoughts do not trip the draft heuristic, so expand is still CoW of every residual and still slower than APC. Second, GSM8K answer-marker stop is a real latency win: $0.33\times$ decode tokens and $0.75\times$ e2e at tp=4 with accuracy 0.925 vs. 0.938. Game24 and HumanEval keep APC-level e2e because those traces do not hit the stop list, while peak KV remains below APC ($0.62\times$ and $0.96\times$). Setting N extends that storage claim to six families at $n=40$: GSM8K peak $0.72\text{--}0.75\times$ APC, Game24 $0.57\text{--}0.62\times$, ForkServe accuracy with APC. The last-line rerun at $n=32$ scores 18/32 on DeepSeek-R1 (APC 16/32), 22/32 on

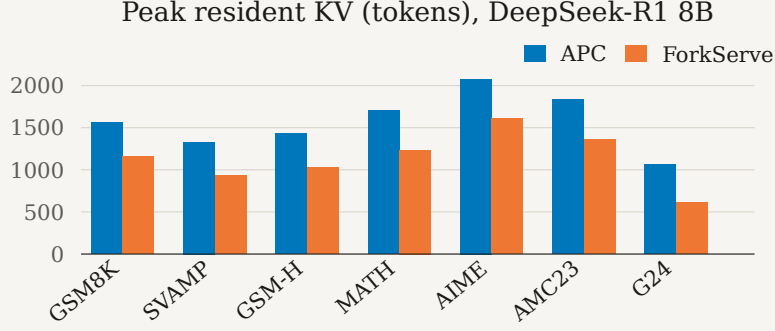


Figure 16: DeepSeek-R1 8B peak KV, shared by settings I and J. Every file is below APC. Game24 is the largest abort gap (0.58 $\times$); AIME the smallest (0.77 $\times$).

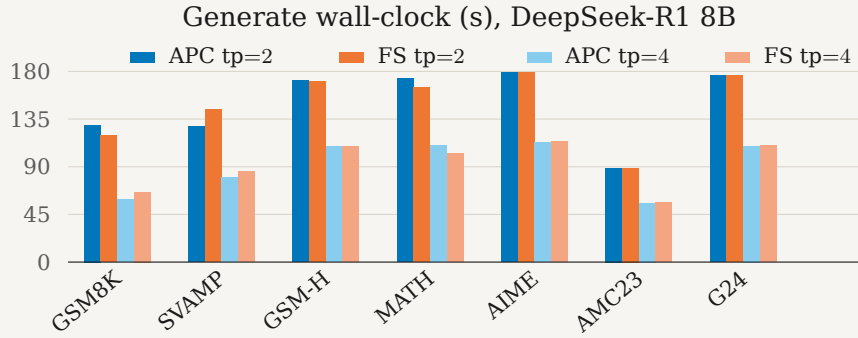


Figure 17: DeepSeek-R1 8B end-to-end generate time. Light bars (tp=4) are 0.56–0.64 $\times$ the dark bars. Orange sits on blue except where winner length differs (GSM8K, SVAMP).

Llama-3-8B (APC 23/32), and 6/32 on Mistral-7B (APC 11/32), with GSM8K peak 0.73–0.74 $\times$ APC. Mistral-7B HumanEval pass@1 is 0.350 (14/40) on recompute, APC, and ForkServe. Setting O repeats the storage claim under a change of k : spine peak is flat and the ratio is 0.53–0.55 at $k=8$ on Llama-3-8B, Mistral-7B, and DeepSeek-R1. Draft prune is what cuts that fan-out (179 vs. 307 ms on Llama-3-8B, 194 vs. 363 ms on DeepSeek-R1); the last-line stop is not, because the winner already ends inside D . Setting P is the same claim at $D=2048$: MATH-500 peak 0.75 $\times$ (DeepSeek-R1) and 0.79 $\times$ (Qwen3-4B). ForkServe+ prunes 75 MATH-500 residuals on both models and returns fan-out to 1553 ms and 1223 ms, against APC at 1604 ms and 1178 ms, with generate time matched to APC.

The remaining full-set runs are Tree-of-Thoughts on Game24 ($n=1362$) and ReAct on HumanEval ($n=164$) for ForkServe; vLLM recompute and APC already finished those files. A live-scheduler concurrent-load experiment should likewise confirm the analytical C and Θ of setting K.

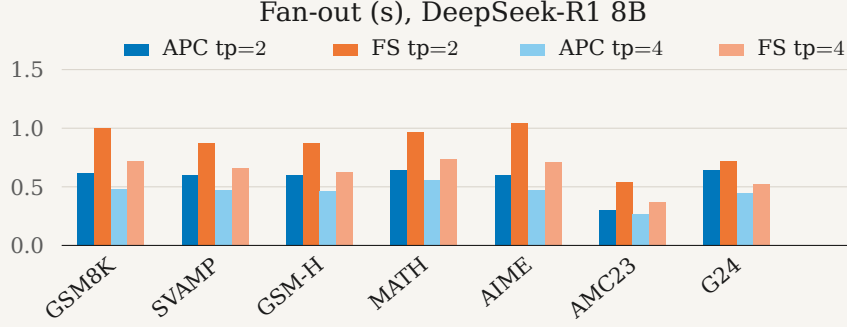


Figure 18: DeepSeek-R1 8B fan-out. The abort/CoW tax is tenths of a second against 56-179s generate, so it does not set T_{e2e} .

Model	tp	APC acc	FS acc	Δ	peak FS/APC	FS e2e vs. APC	4/2
Qwen3-8B	2	0.618 (317)	0.635 (326)	+1.8 pp	0.62-0.80	within 3%	—
Qwen3-8B	4	0.622 (319)	0.608 (312)	-1.4 pp	0.62-0.80	within 6%	0.66
Qwen3-14B	2	0.591 (303)	0.596 (306)	+0.6 pp	0.62-0.80	within 5%	—
Qwen3-14B	4	0.591 (303)	0.577 (296)	-1.4 pp	0.62-0.80	within 2%	0.61-0.64
DeepSeek-R1 8B	2	0.368 (189)	0.388 (199)	+2.0 pp	0.58-0.77	-0.5% pool	—
DeepSeek-R1 8B	4	0.374 (192)	0.396 (203)	+2.1 pp	0.58-0.77	+1.3% pool	0.56-0.64

Table 27: Micro-average accuracy and system summary on the 513-item union. Storage ratios are unchanged by tensor width. Quality stays within a few problems of APC on Qwen and is ahead on DeepSeek-R1. Four-GPU wall-clock is $0.56\text{--}0.66\times$ two GPUs on every checkpoint.

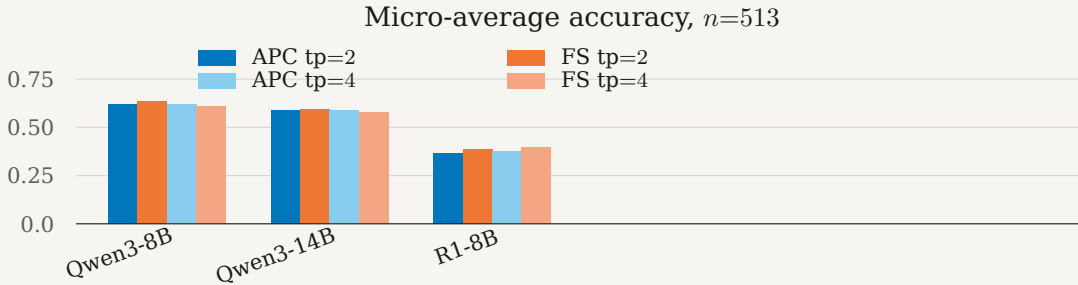


Figure 19: Pool accuracy across checkpoints. Orange sits on blue inside each model; the vertical gap between Qwen and R1 is the checkpoint and $D=2048$, not APC vs. ForkServe.

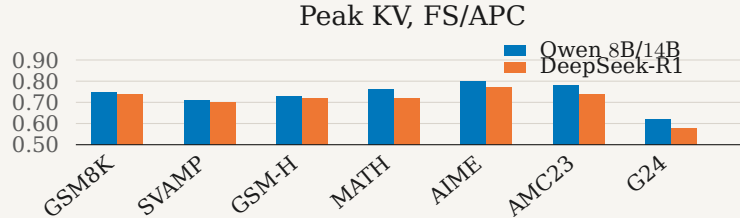


Figure 20: Storage ratio by file. Qwen3-8B and Qwen3-14B share the left bar. Every ratio is below 1; Game24 is the largest abort gap on both families.

	prefilled	pruned	abort-mark (ms)	fan-out (ms)	CoW (ms)	peak KV
ForkServe	4	0	0.22	1.31	1.09	1064
ForkServe+	2	2	0.10	1.65	1.55	1064

Table 28: Mock ToT micro-fan-out (8 sessions, $k=4$). Plus prunes two residuals before GPU prefill; peak KV is still the committed path.

QPS	in-flight	APC run	FS run	APC tok/s	FS+ tok/s	APC p99 (s)	FS+ p99 (s)
8	3	3	3	1920	1920	0.020	0.020
16	6	6	6	3840	3840	0.020	0.020
32	13	13	13	8320	8320	0.020	0.020
48	19	19	19	12160	12160	0.020	0.020
64	26	26	26	16640	16640	0.020	0.020
96	38	37	38	23680	24320	0.042	0.020
128	51	37	50	23680	32000	0.323	0.036
192	77	37	50	23680	32000	0.885	0.452
256	102	37	50	23680	32000	1.425	0.852

Table 29: Analytical QPS sweep from (12) with GSM8K peaks 1563 vs. 1163. APC saturates at 37 slots / 23680 tok/s; ForkServe+ at 50 / 32000 tok/s (+35%). Max QPS with P99 TTFT ≤ 1 s is 192 (APC) vs. 256 (ForkServe+).

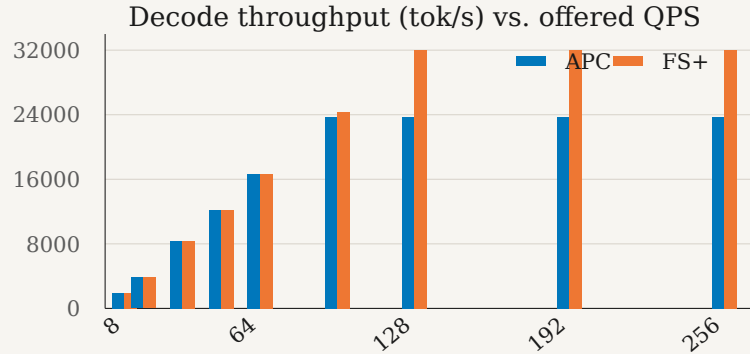


Figure 21: Setting K throughput. The two stacks match until the APC slot cap (37 trees). ForkServe+ continues to 50 trees / 32000 tok/s.

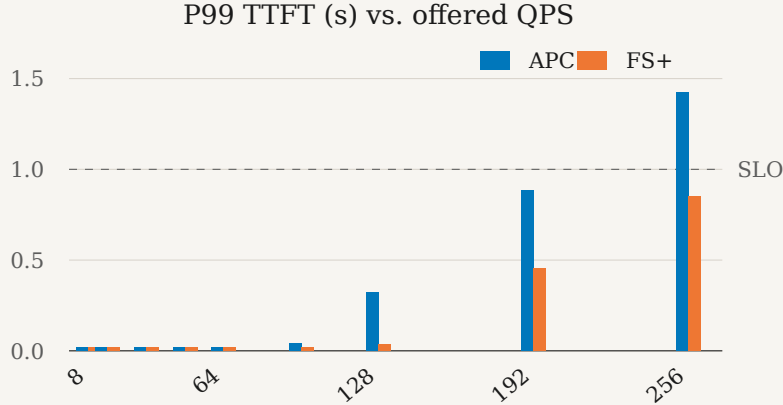


Figure 22: Setting K tail latency. APC crosses the 1 s SLO between 192 and 256 QPS; ForkServe+ remains under the SLO at 256.

Method	prefill tok	prefill ms	xfer tok	fan-out ms	peak KV
APC	11520	138.2	0	139.0	11520
ForkServe	3840	46.1	0	47.1	3840
hash prefill	11520	138.2	0	139.0	11520
disagg. prefill	11520	138.2	11520	162.1	11520
APP	2912	34.9	352	35.9	2880

Table 30: Setting L, first fan-out. Control-plane cost model, not A100 wall-clock. APP prunes 20 residuals across the ten sessions and ships 352 tokens instead of the full clone.

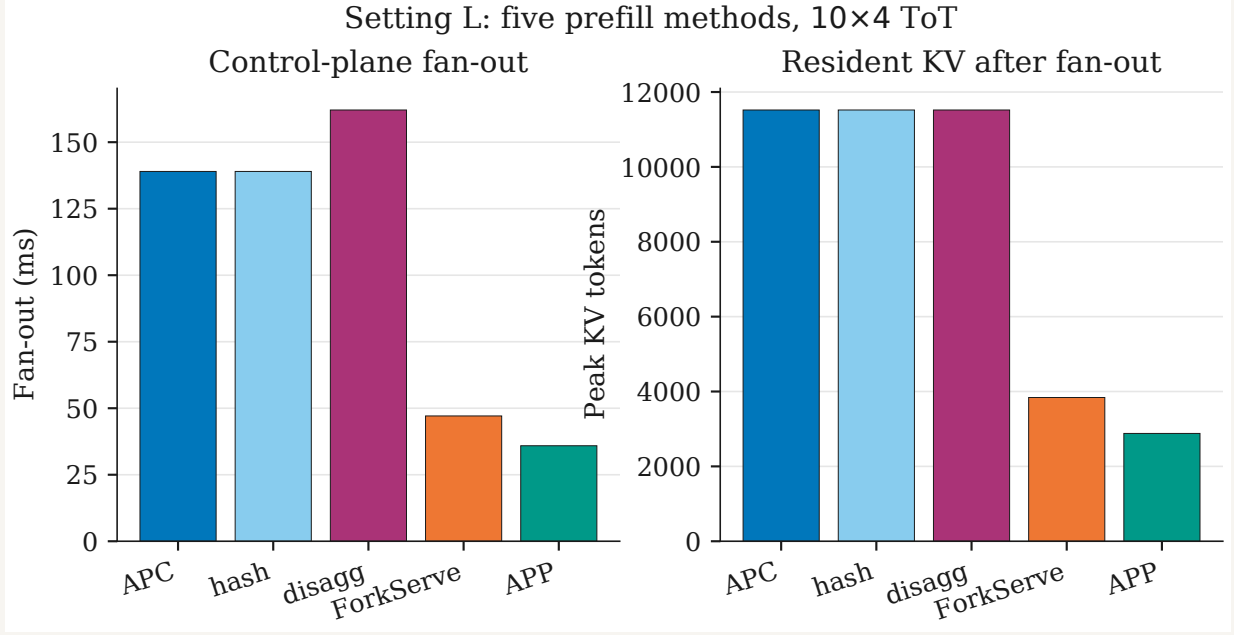


Figure 23: Setting L visualization. Hash prefill matches APC on the first expand. Stock disaggregated prefill is slower because it ships the clone. ForkServe aliases the trunk; APP also drops the two low-score residuals.

tp	System	e2e (s)	fan-out (ms)	peak KV	decode tok	tok/item	acc.
2	recompute	37.34	2006	4744	40960	512	0.925
2	APC	36.00	558	1582	40960	512	0.900
2	ForkServe	36.04	1181	1182	21235	265	0.887
2	ForkServe+	28.22	1261	1182	13356	167	0.863
4	recompute	23.36	1236	4744	40960	512	0.925
4	APC	22.72	438	1582	40960	512	0.938
4	ForkServe	23.30	994	1182	20997	262	0.925
4	ForkServe+	16.94	1048	1182	12826	160	0.925

Table 31: Setting M, GSM8K prefix $n=80$, $D=512$. Peak KV is identical at $tp=2$ and $tp=4$: occupancy tracks the committed spine, not the rank count. ForkServe+ / APC peak = $1182/1582=0.75$. Decode tokens follow (7): APC always spends nD ; ForkServe already hits EOS ($0.52\times$); the stop list cuts that to $0.33\times$ and e2e to $0.78\times$ APC at $tp=2$ and $0.75\times$ at $tp=4$. Accuracy at four GPUs is $74/80$ vs. APC $75/80$.

tp	System	Game24 e2e (s)	peak	decode tok	success	HE e2e (s)	HE peak	HE pass@1
2	recompute	43.97	3068	40960	0.375	45.90	3796	0.425
2	APC	43.14	1289	40960	0.463	44.78	1970	0.438
2	ForkServe	42.98	793	35419	0.438	44.64	1882	0.450
2	ForkServe+	43.05	793	34427	0.350	44.70	1882	0.450
4	recompute	28.97	3068	40960	0.487	30.42	3796	0.450
4	APC	28.51	1289	40960	0.338	29.85	1970	0.425
4	ForkServe	28.36	793	35513	0.463	29.69	1882	0.450
4	ForkServe+	28.40	793	34055	0.375	29.77	1882	0.412

Table 32: Setting M, Game24 and HumanEval, $n=80$. Peak ratios vs. APC: Game24 $793/1289=0.62$, HumanEval $1882/1970=0.96$ (wrap-dominated, $k=2$), matching (8). Decode tokens barely move, so e2e stays within 1% of APC. Game24 success under ForkServe+ is lower than ForkServe at both widths; HumanEval pass@1 is matched at $tp=2$ and -3 problems at $tp=4$.

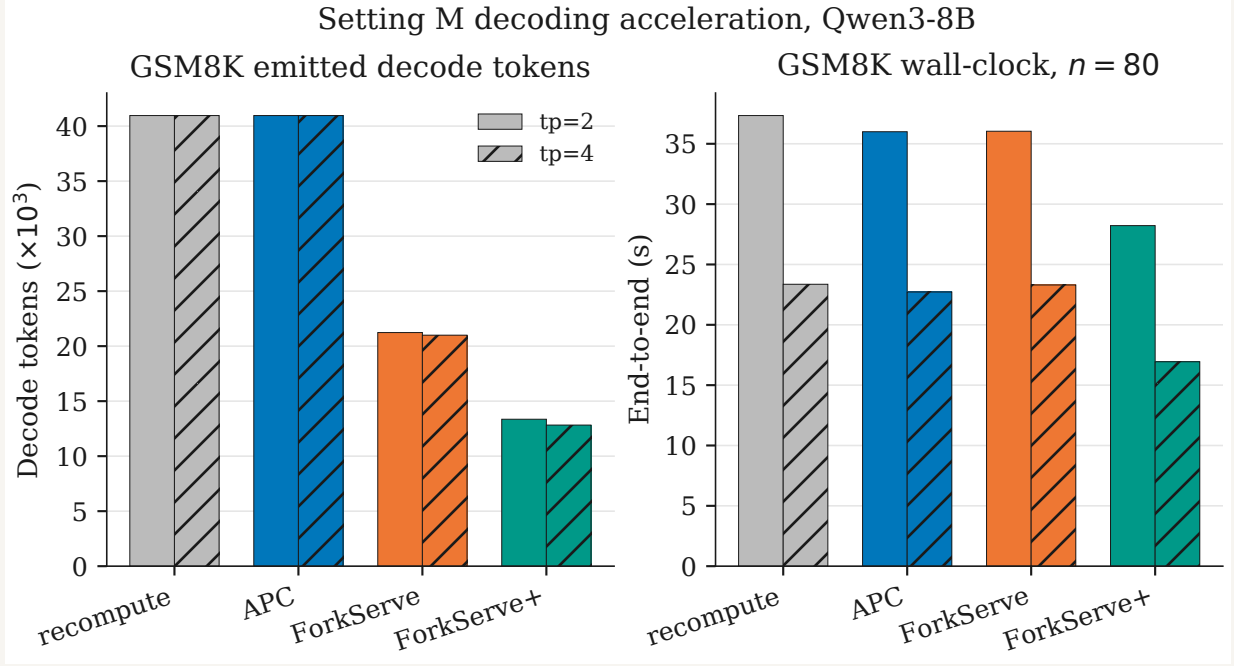


Figure 24: Setting M, GSM8K. Hatched bars are $tp=4$. APC spends the full nD budget; ForkServe already stops on EOS; ForkServe+ stops at #####. End-to-end follows D^+ , not fan-out.

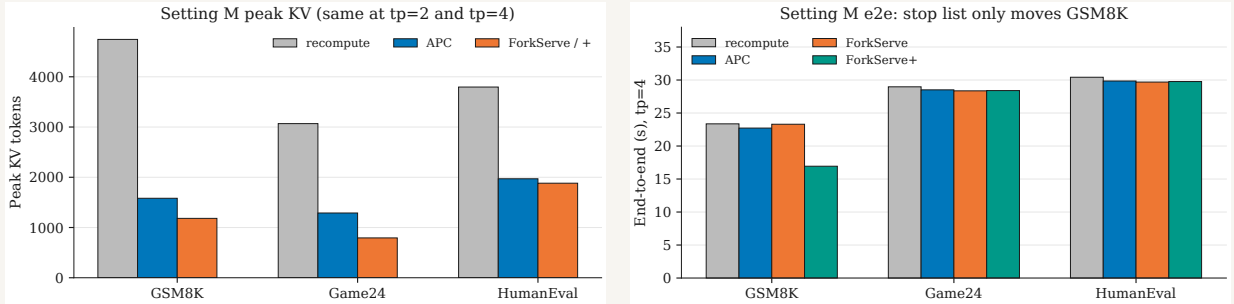


Figure 25: Setting M storage vs. latency. Left: peak KV is identical for ForkServe and ForkServe+, below APC on ToT ($0.75\times$, $0.62\times$) and close on ReAct ($0.96\times$). Right: at $tp=4$, only GSM8K e2e drops, because Game24 and HumanEval do not hit the stop list.

GSM8K, $tp=2$	Qwen3-4B	Math-1.5B	Qwen3-14B	Llama-3-8B	Mistral-7B	R1-8B
APC peak	1512	1512	1512	1509	1569	1405
FS / FS+ peak	1112	1112	1112	1109	1169	1005
ratio vs. APC	0.74	0.74	0.74	0.73	0.75	0.72
APC D^+	20480	20480	20480	20480	20480	20480
FS D^+	9860	17070	12087	5560	10013	14974
FS+ D^+	7206	11704	11162	5262	9660	—
APC acc.	0.925	0.225	0.850	0.625	0.500	0.650
FS acc.	0.875	0.275	0.825	0.675	0.475	0.650
FS+ acc.	0.550	0.275	0.725	0.675	0.350	—
APC e2e (s)	14.00	8.31	35.92	11.44	16.14	21.43
FS e2e (s)	14.28	8.63	36.20	11.67	18.33	21.74
FS+ e2e (s)	12.56	8.63	36.10	11.07	18.10	—

Table 33: Setting N, GSM8K $n=40$, $tp=2$. Peak KV is $0.72\text{--}0.75\times$ APC on every family and does not change at $tp=4$. ForkServe accuracy stays with APC (± 2 problems except Math-1.5B, where both are weak). ForkServe+ stop is safe on Llama-3-8B and cuts tokens on Qwen without always matching accuracy.

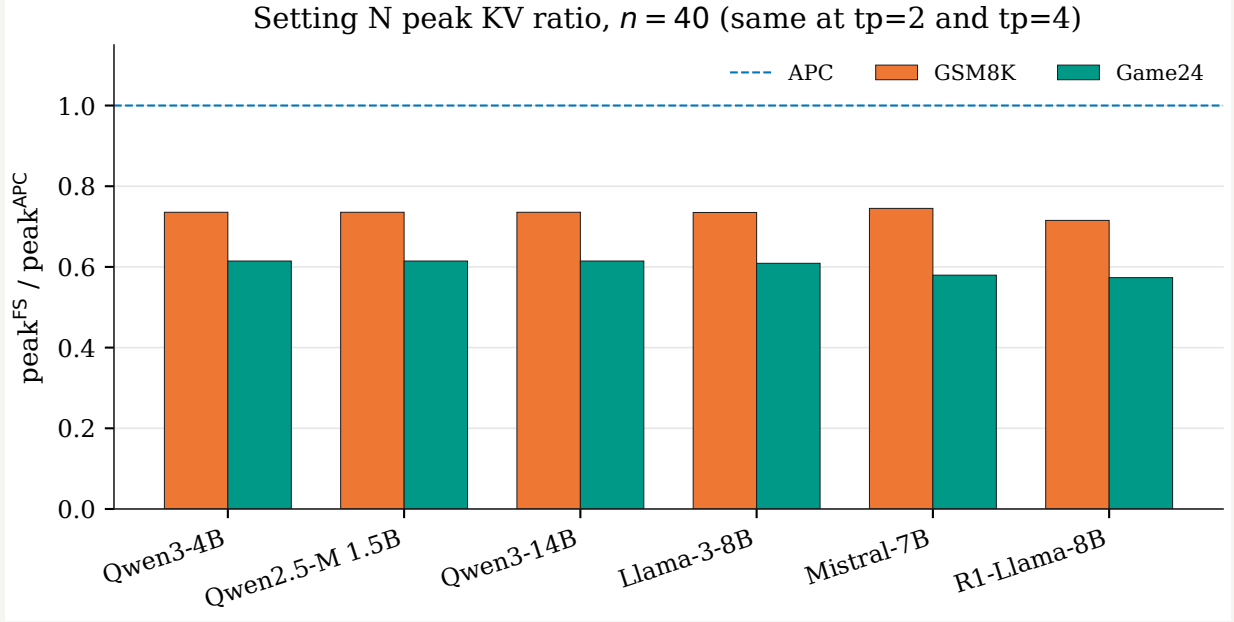


Figure 26: Setting N peak-KV ratio vs. APC. GSM8K sits in $[0.72, 0.75]$; Game24 in $[0.57, 0.62]$. The abort gap of (8) is a property of k and residual length, not of the weight family.

Game24 peak (any tp)	APC	FS / FS+	ratio
Qwen3-4B / Math-1.5B / Qwen3-14B	1287	791	0.61
Llama-3-8B	1248	760	0.61
Mistral-7B	1351	783	0.58
DeepSeek-R1-8B	1144	656	0.57

Table 34: Game24 peak KV. Short trunks make unused residuals a larger fraction of M_{APC} , so the ratio is below GSM8K, as (8) predicts.

HumanEval pass@1, $n=40$, $tp=2$	Llama-3-8B	Mistral-7B	Qwen3-14B
recompute	0.550	0.350	0.525
APC	0.500	0.350	0.575
FS / FS+	0.475	0.350	0.550
peak FS / APC	0.90	0.89	0.95
APC e2e (s)	12.75	11.69	37.71
FS e2e (s)	14.93	13.77	37.55

Table 35: Setting N HumanEval. Mistral pass@1 is 0.350 on every system (same at $tp=4$). Peak ratios stay wrap-dominated ($k=2$). At $tp=4$, Llama is 0.600 / 0.600 / 0.575 and Qwen3-14B is 0.525 / 0.575 / 0.575 (recompute / APC / FS).

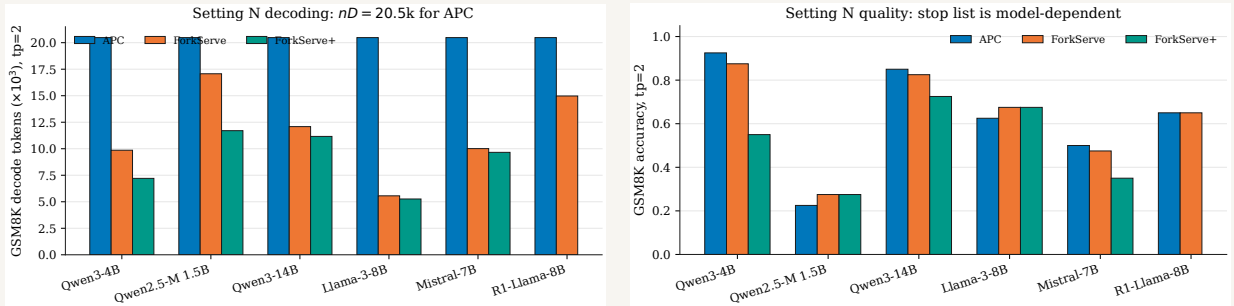


Figure 27: Setting N GSM8K, $tp=2$. Left: APC always emits nD ; ForkServe already stops on EOS; ForkServe+ stops earlier. Right: ForkServe tracks APC accuracy; ForkServe+ matches on Llama-3-8B and drops on Qwen3-4B / Mistral / Qwen3-14B.

		DeepSeek-R1-8B			Llama-3-8B			Mistral-7B		
		APC	FS	FS+	APC	FS	FS+	APC	FS	FS+
GSM8K	peak	1493	1093	1093	1485	1085	1085	1521	1121	1121
	accuracy	0.500	0.562	0.562	0.719	0.750	0.688	0.344	0.344	0.188
	correct	16/32	18/32	18/32	23/32	24/32	22/32	11/32	11/32	6/32
	decode tokens	16384	13887	12693	16384	4711	5101	16384	6999	6723
	e2e (s)	17.64	17.66	17.61	12.87	10.13	12.98	11.27	13.88	13.46
Game24	peak	1064	616	616	1224	736	736	1296	728	728
	success	1/32	0/32	1/32	0/32	0/32	0/32	0/32	0/32	0/32
	e2e (s)	9.00	8.91	8.82	0.50	0.43	0.32	8.57	8.55	8.36

Table 36: Last-line answer stop, $n=32$, $tp=2$. GSM8K peak is $1093/1493=0.73$, $1085/1485=0.73$, and $1121/1521=0.74$ of APC. ForkServe+ decode tokens are $0.77\times$, $0.31\times$, and $0.41\times$ APC. DeepSeek-R1 ForkServe+ scores 18/32, above APC. Llama-3-8B stays within one problem of APC; Mistral-7B loses five problems. Game24 peaks are $0.58\times$, $0.60\times$, and $0.56\times$ APC. Success stays at most 1/32 because $D=256$ does not finish an equation. Llama-3-8B Game24 emits 32 decode tokens on every system (immediate EOS).

		Llama-3-8B				Mistral-7B			
		APC	FS	FS+	FS/APC	APC	FS	FS+	FS/APC
$k=2$	peak	610	550	550	0.90	629	569	569	0.90
	accuracy	12/16	10/16	11/16		4/16	4/16	5/16	
	fan-out (ms)	129	181	174		133	190	177	
$k=4$	peak	750	550	550	0.73	769	569	569	0.74
	accuracy	10/16	9/16	9/16		5/16	5/16	5/16	
	fan-out (ms)	155	238	176		169	247	175	
$k=8$	peak	998	550	550	0.55	1033	569	569	0.55
	accuracy	10/16	11/16	10/16		6/16	4/16	4/16	
	fan-out (ms)	224	307	179		258	349	191	
depth 2	peak	814	614	614	0.75	841	641	641	0.76
	accuracy	8/16	8/16	8/16		6/16	6/16	6/16	

Table 37: Setting O, GSM8K $n=16$, $tp=2$, $D=512$. The spine peak is independent of k ; APC peak grows with the hashed union. DeepSeek-R1 at the same n is $561/629=0.89$ at $k=2$, $561/761=0.74$ at $k=4$, $561/1053=0.53$ at $k=8$, and $641/841=0.76$ at depth 2. ForkServe+ accuracy is 8/16, 8/16, 10/16, and 1/16 against APC 10/16, 9/16, 8/16, and 1/16.

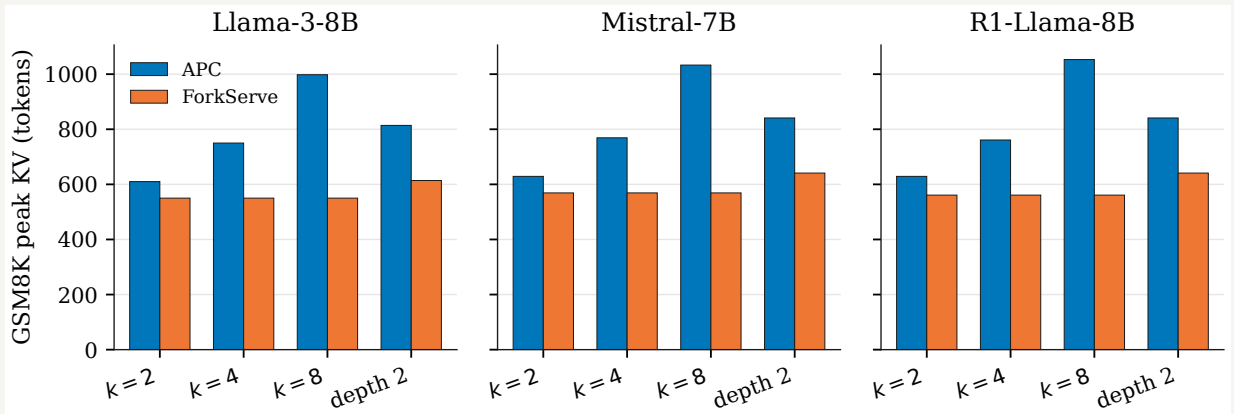


Figure 28: Setting O peak KV. ForkServe+ matches the ForkServe bar on every cell. The spine stays at 550, 569, and 561 from $k=2$ to $k=8$.

		Llama-3-8B		Mistral-7B		DeepSeek-R1	
		APC	FS+	APC	FS+	APC	FS+
1×16	peak	226	176	237	187	246	196
	e2e (s)	19.8	21.9	28.2	27.8	52.9	52.4
	accuracy	10/16	9/16	4/16	5/16	8/16	7/16
16 in one step	peak	2871	2071	2935	2135	2893	2093
	e2e (s)	2.4	4.4	4.4	3.6	4.7	4.7
	accuracy	12/16	10/16	4/16	6/16	10/16	7/16

Table 38: Setting O batch width, GSM8K $n=16$, $k=4$. Peak ratio is 0.78-0.80 at one item, and 0.72-0.73 when all 16 share a step. The wide step is the first GPU measurement of the concurrency term in (12); it is not a closed-loop QPS sweep.

		MATH-500 ($n=200$)			AIME ($n=73$)		
		APC	FS	FS+	APC	FS	FS+
DeepSeek-R1	peak	1863	1391	1391	2118	1646	1646
	accuracy	111/200	101/200	101/200	5/73	7/73	4/73
	decode tokens	409600	227470	228991	149504	146008	145879
	e2e (s)	432.7	431.2	433.2	178.3	178.3	199.9
	fan-out (ms)	1604	2375	1553	613	988	670
Qwen3-4B	peak	1937	1521	1521	2150	1734	1734
	accuracy	127/200	123/200	127/200	1/73	5/73	4/73
	decode tokens	409600	286935	265299	149504	147851	148682
	e2e (s)	328.3	326.3	322.2	135.4	135.3	138.0
	fan-out (ms)	1178	1750	1223	462	748	501

Table 39: Setting P, $D=2048$, $k=4$, $tp=2$, batches of eight. Peak ratios are 0.75 and 0.78 on DeepSeek-R1, 0.79 and 0.81 on Qwen3-4B. ForkServe+ prunes 75 MATH-500 branches and 30 AIME branches on both models, which brings MATH-500 fan-out to 1553ms (APC 1604) and 1223ms (APC 1178). Qwen3-4B ForkServe+ matches the APC MATH-500 score (127/200). End-to-end stays with APC except DeepSeek-R1 AIME ForkServe+ (199.9 vs. 178.3s).

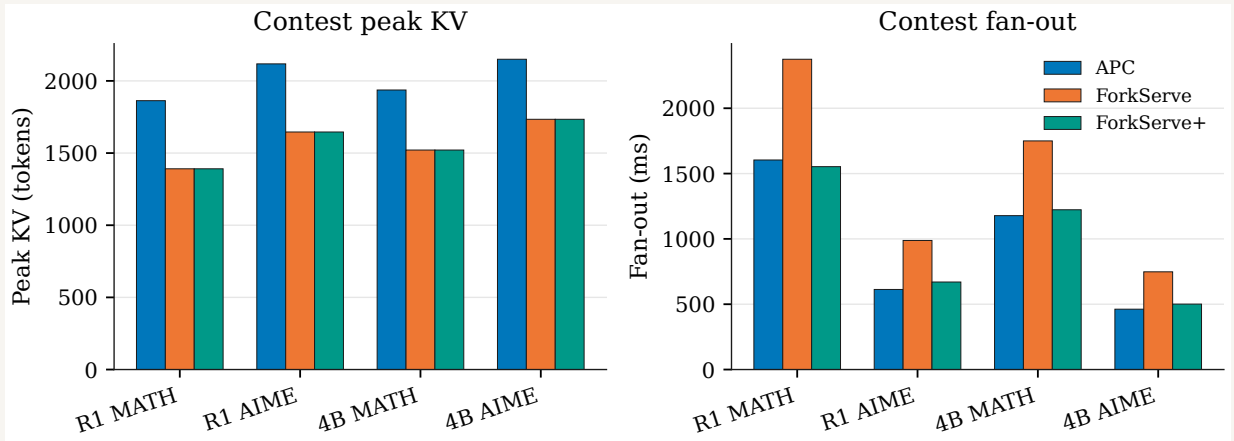


Figure 29: Setting P peak KV and fan-out. ForkServe+ keeps the spine peak and closes the fan-out gap that ForkServe leaves against APC.